



Machine Learning

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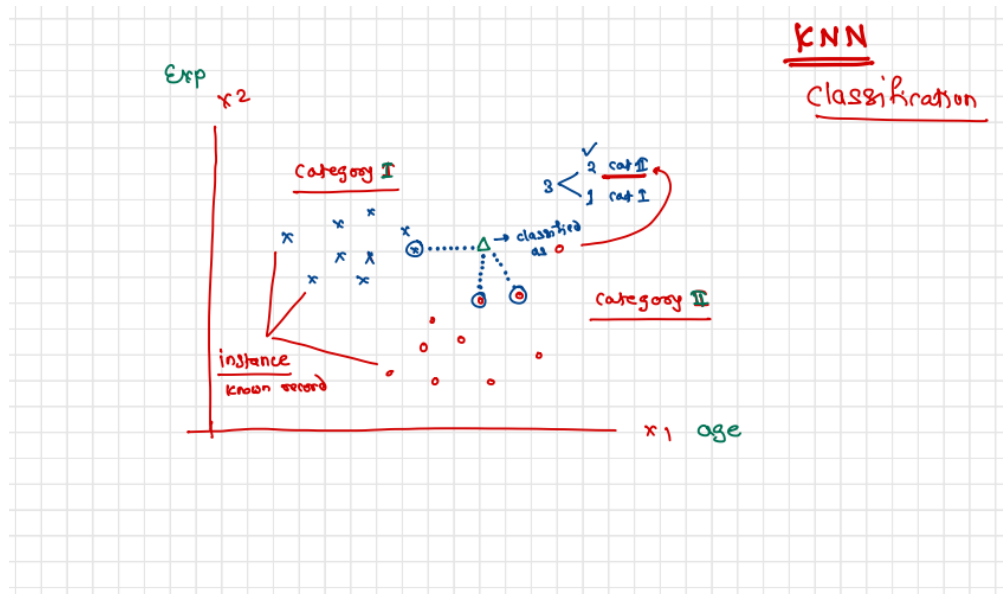
1. What is Machine Learning?

- A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .
- Machine Learning is the science (and art) of programming computers so they can learn from data.
- Past Data $\rightarrow$ Analyse $\rightarrow$ Model $\rightarrow$ Prediction

2. Types of Machine Learning

- ▼ Whether or not they are trained with human supervision
 - ▼ Supervised Learning
 - The majority of practical machine learning uses Supervised Learning.

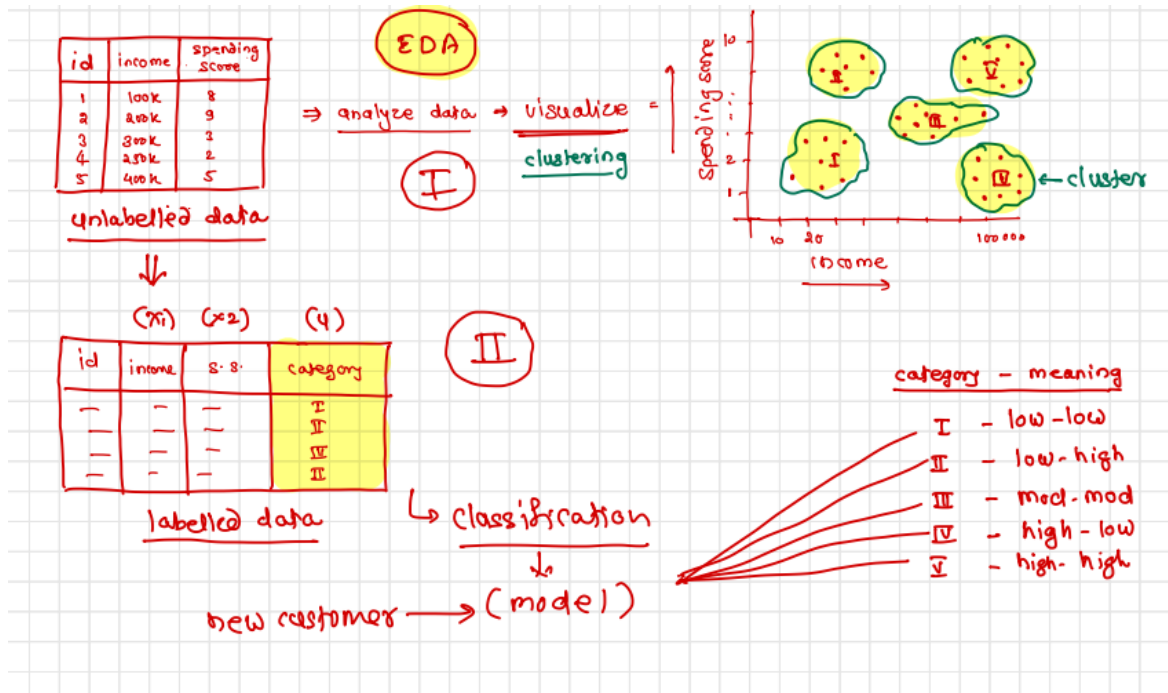
- Input Variable $\rightarrow x$
Output Variable $\rightarrow y$
So,
 $y = f(x)$
- This means, the data we need must be **Labelled**, which means, the data must have independent variables (x) as well as dependent variables (y).
- It is called Supervised Learning because the process of an algorithm learning from the training data-set can be thought of as a teacher supervising the learning process. Here, teacher $\rightarrow$ model evaluation metrics
- As we know the correct answers (observed/dependant values), the model keeps making predictions on the training data and the mistakes are corrected by the teacher (model evaluation metrics).
- Learning stops when the algorithm reaches an acceptable level of accuracy.
- Supervised Learning Problems:
 - Regression:
 - Related to predicting future values (discrete or continuous values).
 - Algorithms:
 - Linear and Multi-Linear Regression
 - Logistic Regression
 - Naive Bayes
 - Support Vector Machine
 - Classification
 - Related to classify the records based on the categories found in dependent variable.
 - Kind of Regression
 - Here, we have:
 - Labelled Data
 - Categorical Dependent Variable (Y)
 - Categories in Y are also known as :
 - Classes
 - Labels
 - Algorithms:
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - Support Vector Machine
 - K-Nearest Neighbour



▼ Unsupervised Learning

- Unsupervised Learning is where you only have input data (X) and no corresponding output variable.
- We use it for **Unlabelled** Data. This means, output/dependant variable (Y) is absent while independent variable (X) is present.
- Used for Exploratory Data Analysis (EDA) and not Prediction, which basically means that we get insight on the already available data.
- The goal for Unsupervised Learning is to model the underlying structure or distribution in the data in order to learn more about the data.
- Unlike Supervised Learning, we do not have a teacher (model) and correct answers (Y) here.
- Unlabelled Data → Unsupervised Learning → Labelled Data → Supervised Learning(Optional)
- Unsupervised Learning Problems:
 - Clustering:
 - Discover the inherent grouping in the data, such as grouping (complete)
 - Algorithms:
 - K-Means Clustering
 - Hierarchical Clustering
 - Association Rules Mining (Market Basket Analysis):
 - An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y.
 - Algorithms
 - Apriori

- Eclat



▼ Reinforcement Learning

- Agent Based Training → Experience Based Learning
- The training is based on positive and negative feedback/reward.
- So, the goal of the machine is to maximise the reward.
- Reinforcement learning differs from the supervised learning in a way that in supervised learning the training data has the answer key with it so the model is trained with the correct answer itself whereas in reinforcement learning, there is no answer but the reinforcement agent decides what to do to perform the given task.
- In absence of training data-set, it is bound to learn from its experience.
- Algorithms:
 - Q-Learning
 - Deep Q-Learning

▼ Whether or not they can learn incrementally on the fly

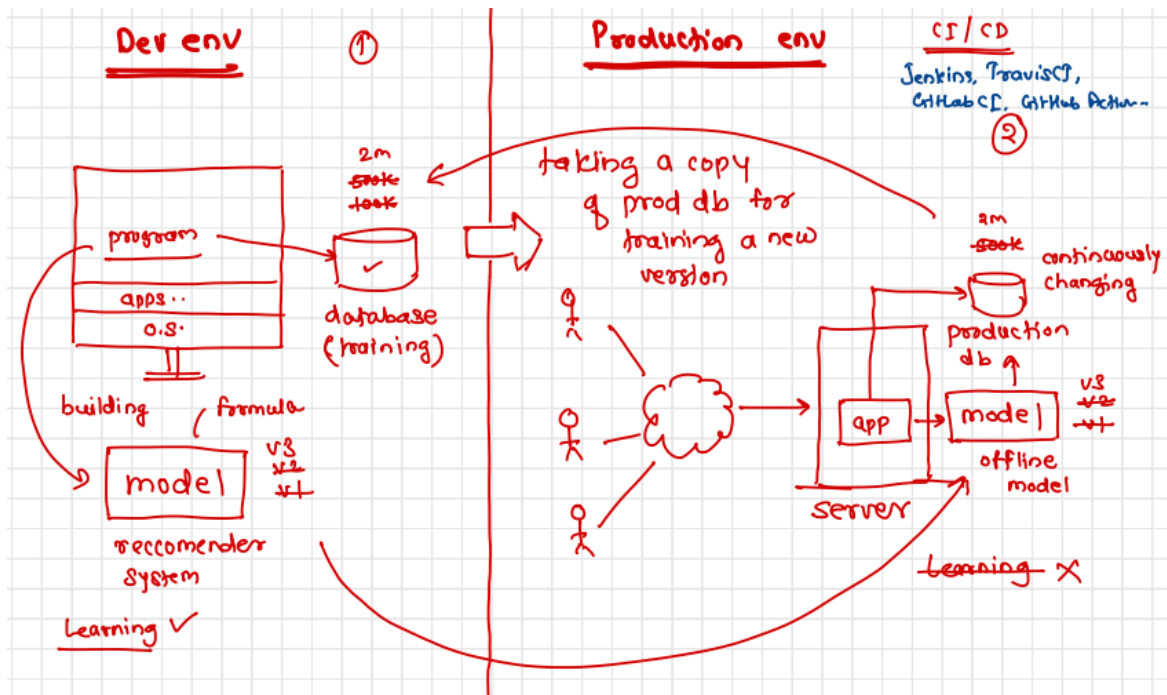
▼ Online Learning

- In Online Learning, you train the system incrementally by feeding it data instances sequentially, either individually or in small groups called **mini-batches**.
- Each learning step is fast and cheap, so the system can learn about new data on the fly, as it arrives → Artificial Neural Network (ANN).

- Online learning is great for systems that receive data as continuous flow (E.x. stock prices) and need to adapt to change rapidly or autonomously.
- It is a good option if you have limited computing resources.
- There are algorithms can also be used to train on a data-set that is not on the machine which is called **Out-Of-Core-Learning**.
- One important parameter of online learning system is how fast they should adapt to changing data: this is called the Learning Rate.
- If the learning rate is high, your system will rapidly adapt to new data, but it will also tend to quickly forget the old data.
- If you set a low learning rate, the system will have more inertia that is, it will learn more slowly, but it will also be less sensitive to noise in the new data or to sequences of non-representative data points (outliers).

▼ Offline/Batch Learning

- In Batch Learning, the system is incapable of learning incrementally.
- It must be trained using all the available data.
- This will generally take a lot of time and computing resources, so, it is typically done offline.
- First the system is trained, and then it is launched into production and runs without learning anymore, it just applies what it has learned
- Disadvantages:
 - If you want a batch learning system to know about the new data-set, you need to train a new version of the system from scratch on the full data-set, then stop the old system and replace it with the new one.
 - The whole process of training, evaluating, and launching a Machine Learning system can be automated easily using DevOps Tools → CI/CD Pipeline.
 - Training using the full set of data can take many hours.
 - Training on the full set of data requires a lot of computing resources (CPU, memory, disk space, disk I/O, network I/O)



▼ Whether they work by simply comparing new data points to known data points, or instead by detecting patterns in the training data and building a predictive model, much like scientists do

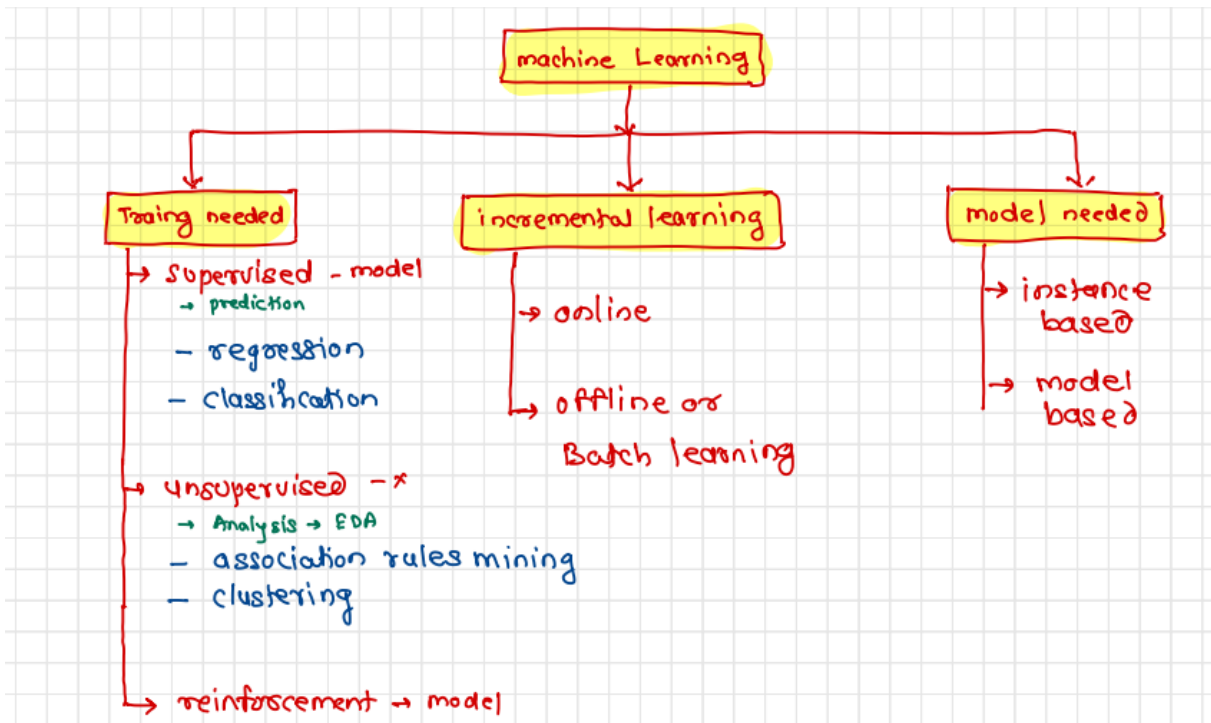
▼ Instance Based

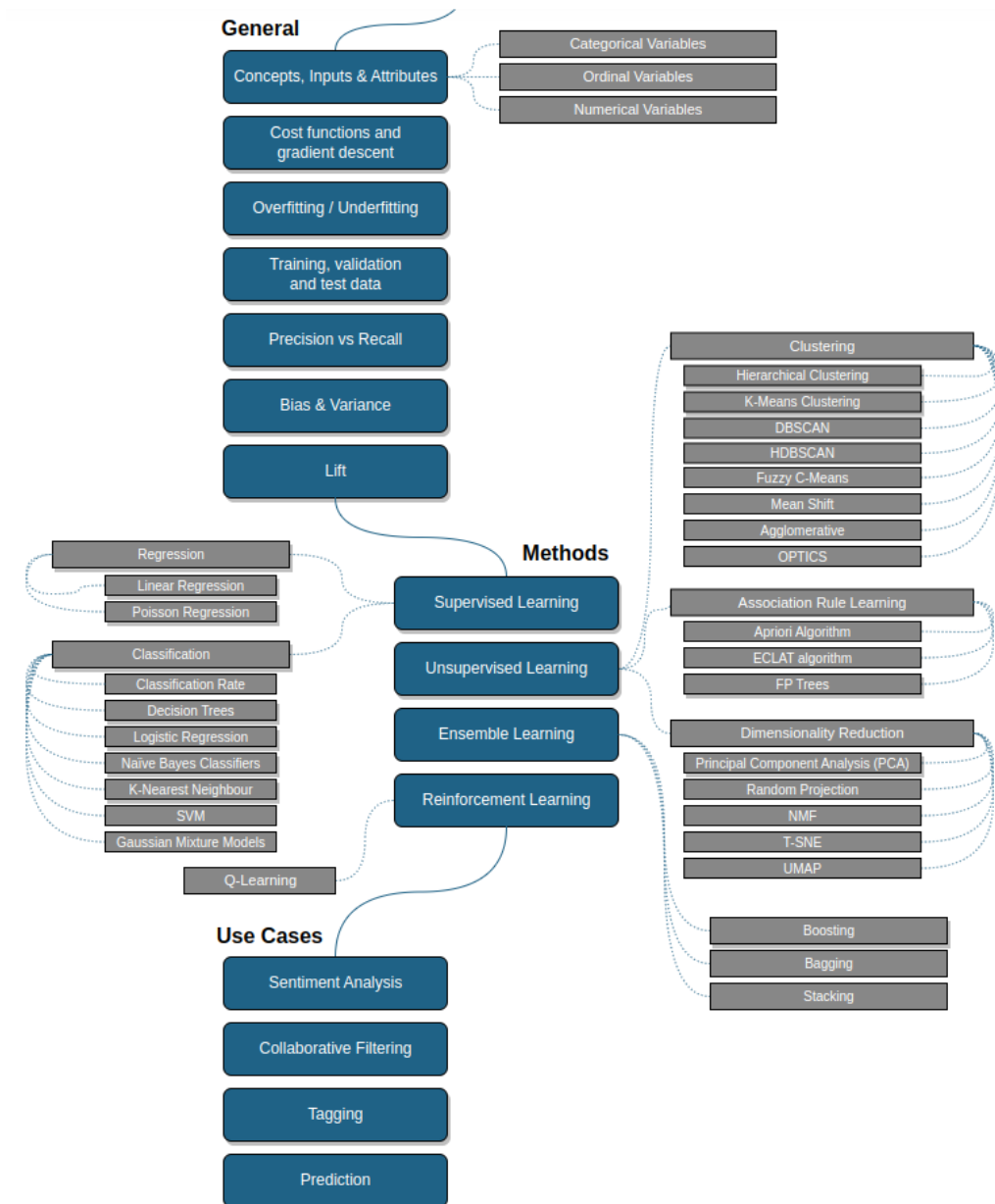
- The system learns the examples by heart.
- Then generalises to new cases by using a similarity measure to compare them to the learned examples (or subset of them).
- It is called instance-based because it builds the hypothesis from the training instances.
- It is also known as **Memory-Based Learning** or **Lazy-Learning** as no model is generated.
- Advantages:
 - Instead of estimation for the entire instance set, local approximations can be made to the target function.
 - This algorithm can adapt to new data easily, one which is collected as we go.
- Disadvantages:
 - Classification costs are high → more resources are required.
 - Large amount of memory required to store the data, and each query involves starting the identification of a local model from scratch.
- Example: KNN (K-Nearest Neighbour)

▼ Model Based

- Train model from training data to estimate model parameters i.e. discover patterns.

- Store the built model in suitable format.
- Generalise the rules of model.
- Predict the unseen instance (data) using the model → consume/use the model.
- It requires a known model form.
- It takes less memory compared to the instance based learning.





3. Supervised Learning

- Supervised learning is a type of machine learning where the algorithm learns from labelled data to make predictions (continuous) or classifications (categorical) on new, unseen data.
- In Supervised Learning, a model is trained on a data-set where the input data (independent variables) and their corresponding output or labels (dependent variables) are known.
- The goal is to learn a mapping function that can predict the output for new input data.
- It can be done using 2 ways:
 - Regression

- Classification

▼ 3.1. Regression

Overview

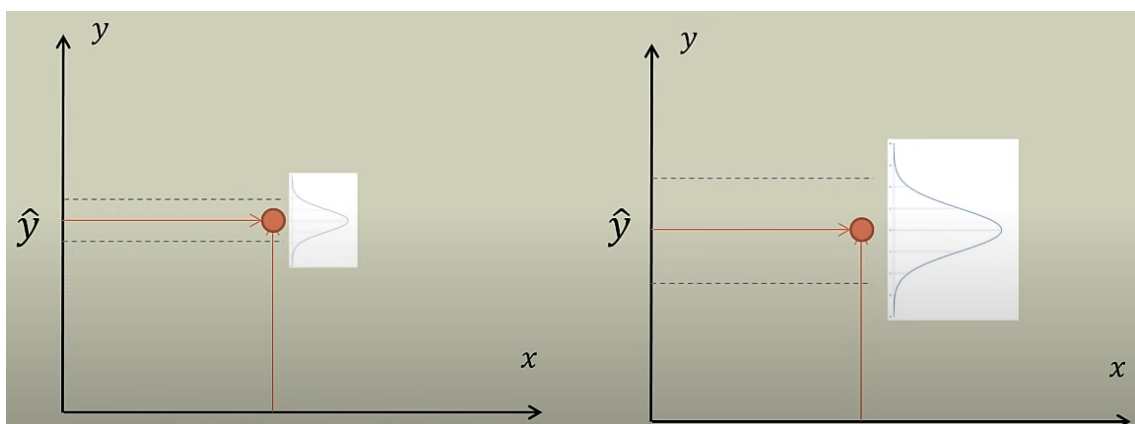
- Regression is a type of supervised machine learning algorithm used to predict a continuous target variable (also known as dependent variable) based on one or more independent variables (also known as features or predictors).
- The goal of regression is to build a mathematical model that can estimate the relationship between the dependent variable and the independent variables, which can be used to make predictions on new data.

Types of Regression

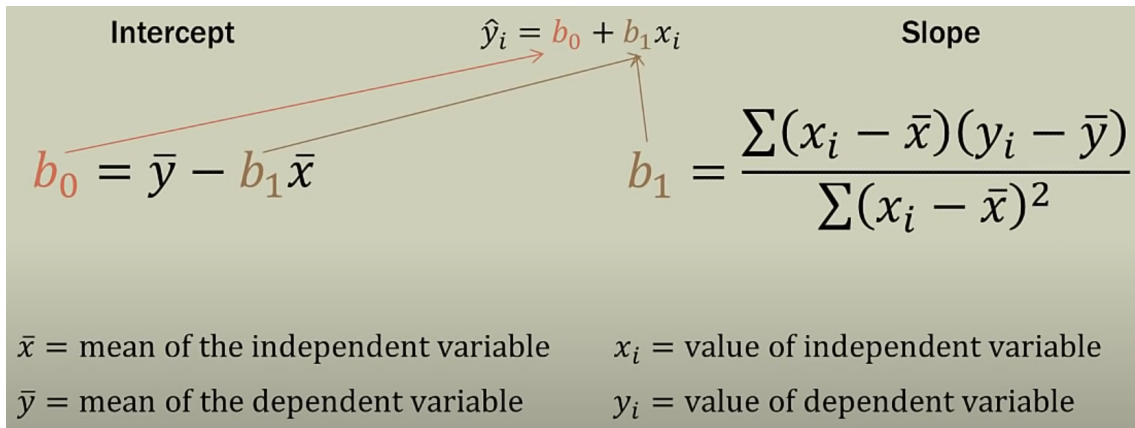
▼ Linear Regression

▼ Overview

- Linear Regression is a type of supervised learning algorithm the goal of which is to find the best-fit straight line that describes the relationship between the input variables (also called independent variables or features) and the output variable (also called dependent variable).
- Linear regression can be used for both simple linear regression (one input variable) and multiple linear regression (more than one input variable).
- In Simple Linear Regression, the dependent variable is not really a point but is actually mean of a distribution of dependent variable values for a given value of the independent variable. This distribution of values of the dependent variable based on the given independent variable value can either be narrow or be wide. ($\hat{y}$, which is called as a Point Estimate or Predicted Output, is a mean value of y for a given value of x)

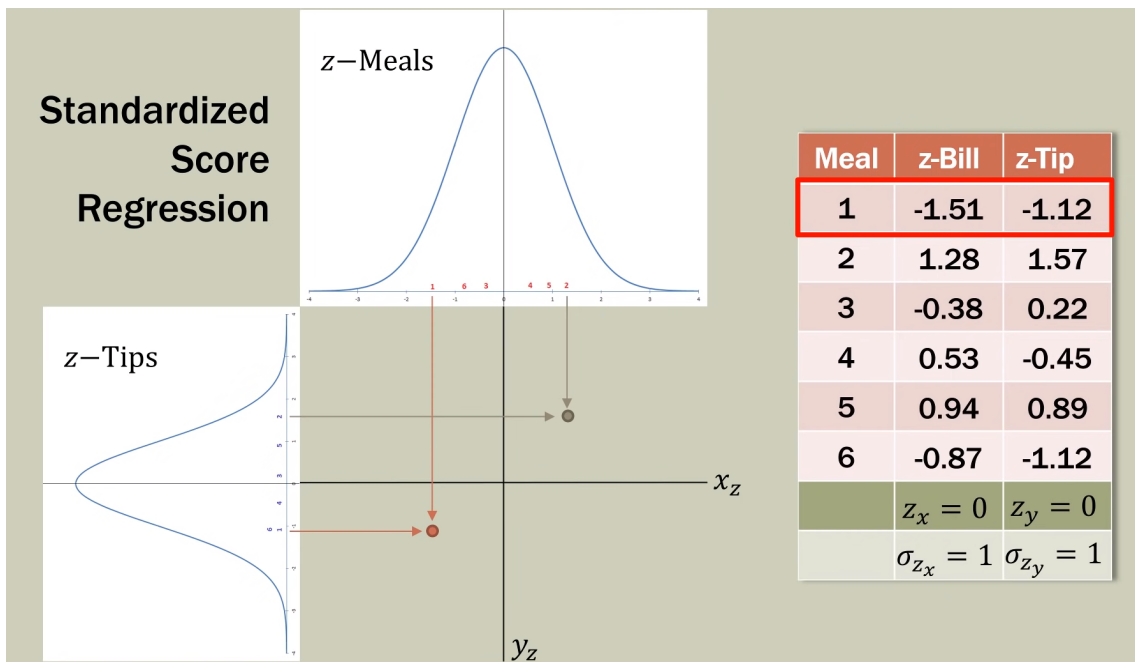


$\hat{y}$ is a Point Estimate of distribution of y 's.

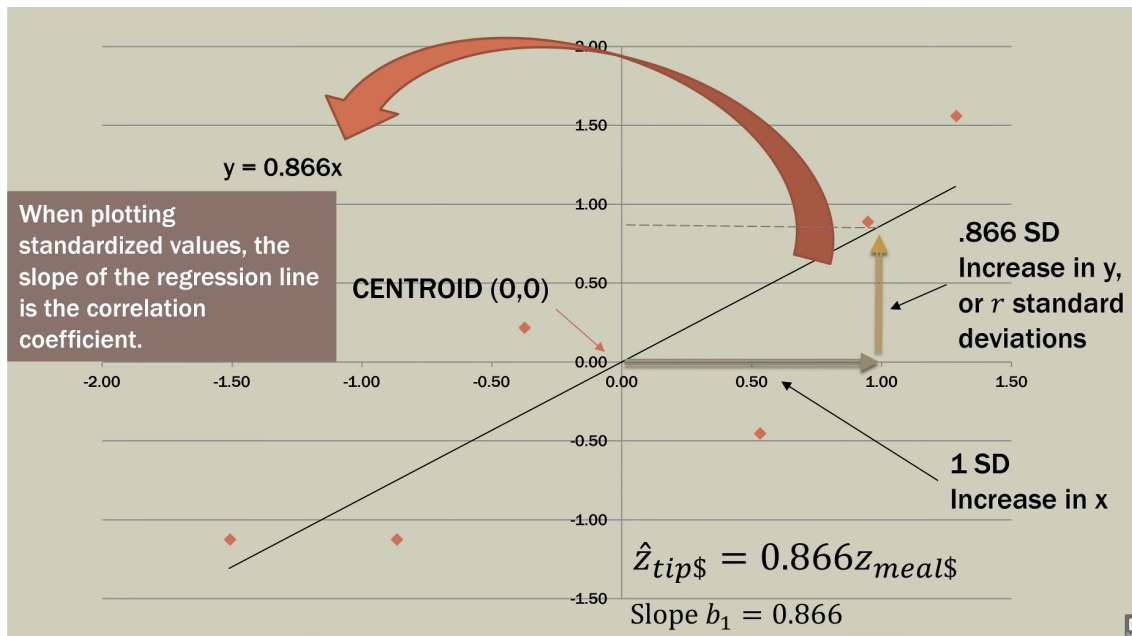


Formulae to calculate the Bi-Variate Linear Regression equation.

- Correlation of dependent and independent variables will be the same even if we standardise the values of both variables (using the z-score formula). This means that Correlation is a standardised comparison parameter.
- After standardising both variables, if we plot them on graph with independent on x axis and dependent on y-axis, we can see that the slope of the resulting line is the correlation between the two variables as shown in below 2 images.



Standardised Bills (z-Meals) vs Standardised Tips (z-Tips)



Detailed Graph.



Out of the total sum of squares, SSR ($\sum((\text{Predicted}-\text{Mean})^2)$) is explained by our Regression Line while the remaining is sum of squared errors (residuals).

▼ How does it work?

- The equation for the line can be expressed as:

$$y = b_0 + b_1 * x$$

- where:
 - y represents the dependent variable
 - b_0 is the y-intercept, which is the value of y when $x = 0$
 - b_1 is the slope of the line, indicating the change in y for a one-unit change in x
 - x is the independent variable
- To determine the line that best fits the data, the coefficients b_0 and b_1 need to be estimated. This is typically done using the method of Ordinary Least Squares (OLS). The OLS method aims to minimise the sum of the squared differences between the observed y -values and the predicted y -values from the regression equation.
- This is achieved by solving a system of equations derived from the OLS method. The equations involve taking partial derivatives and setting them equal to zero.
- The slope b_1 represents the change in the dependent variable y for a one-unit change in the independent variable x . The y-intercept b_0 represents the value

of y when x is equal to zero.

- In multiple linear regression, where there are more than one independent variables, the equation for the line expands to include additional terms for each independent variable, along with their corresponding coefficients. The estimation process involves estimating the coefficients for each independent variable, considering their individual contributions to the dependent variable while accounting for the presence of other variables in the model.

▼ Application

▼ Advantages

- Simplicity and interpretability
- Prediction and forecasting capabilities
- Feature importance and variable selection
- Statistical inference and hypothesis testing
- Complementary with other techniques
- Provides a baseline model for comparison
- Versatile and widely applicable

▼ Disadvantages

- Assumes a linear relationship between variables, which may not hold in all cases.
- Limited flexibility in capturing complex relationships compared to more advanced models.
- Sensitive to outliers and influential data points.
- Requires the independence of errors and constant variance assumptions.
- Not suitable for categorical or non-linear relationships without transformations.
- Prone to overfitting when dealing with a large number of predictors.
- Reliance on the quality and representativeness of the data for accurate results.



To see how the line is fit to the data, click [here](#).

▼ Polynomial Regression

▼ Overview

- Polynomial Regression is a form of linear Regression where, only due to the non-linear relationship between dependent and independent variables, we add some polynomial terms to the Linear Regression to convert it into Polynomial Regression.
- Before feeding the data into a model, in pre-processing stage, we convert the input variables into polynomial terms using some degree.

- The degree of order that we need to use, which is a hyper parameter, we need to choose it wisely (by trial and error) as a higher degree tends to overfit the model while a lower degree underfits the model.

▼ How does it work?

- The equation for the line with degree 2 and with 2 independent variables can be expressed as:

$$y = b_0 + b_1(x_1^1) + b_2(x_1^2) + b_3(x_2^1) + b_4(x_2^2)$$

- This is also a Linear Regression as, like Linear Regression, even here we are trying to find out the relationship between the dependent variable and the coefficients.
- The only difference here is have for every independent variable, there are 1 or more than 1 coefficients.
- Estimate the coefficients ($b_0, b_1, b_2, \dots, b_n$) that minimise the differences between the observed values of the dependent variable and the predicted values from the polynomial regression equation. This can be done using methods such as Ordinary Least Squares (OLS) or Maximum Likelihood Estimation (MLE).

▼ Application

- Modeling physical phenomena, such as projectile motion or gravitational forces, in physics.
- Predicting sales or demand for products based on factors like price, advertising, and competition.
- Studying the impact of environmental factors on species distribution and biodiversity.
- Analyzing the relationship between variables in engineering and optimization problems.

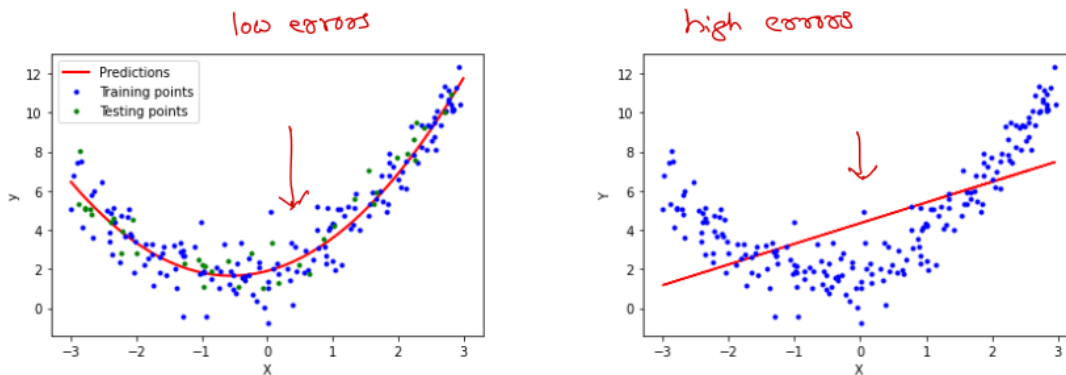
▼ Advantages

- Can capture non-linear relationships between variables.
- Provides flexibility in modeling complex data patterns.
- Allows for more accurate predictions compared to linear regression in non-linear scenarios.
- Offers interpretability, similar to linear regression, with coefficients representing the impact of variables.
- Can serve as a useful exploratory tool to identify non-linear trends in data.

▼ Disadvantages

- Susceptible to overfitting with high-degree polynomials, leading to poor generalization on new data.
- Limited interpretability with higher-degree polynomials, making it challenging to understand the precise functional form of the relationship.
- Increased complexity and computational requirements as the degree of the polynomial increases.

- Vulnerable to the presence of outliers, which can significantly impact the model fit.
- Requires careful selection of the degree of the polynomial, striking a balance between model complexity and avoiding underfitting or overfitting.
- Assumption of a polynomial relationship may not always be appropriate for the underlying data.



▼ Stepwise Regression

▼ Overview

- Stepwise regression is a variable selection procedure for independent variables.
- It consists of a series of steps designed to find the most useful x-variables to include in a regression model.
- At each step of procedure, each variable is evaluated using a set criterion to see if it should remain in the model or not.
- Basis for selection could be:
 - Choosing variable that satisfies the stipulated criterion.
 - Removing the variable that least satisfies the criterion like T-Statistic Value.

▼ Working

- Forward Selection
 - Begins with no variables in the model.
 - Tests each variable as it is added to the model.
 - Then keeps those that are deemed most statistically significant.
 - Repeating the process until the results are optimal.
 - In forward selection, the algorithm starts with an empty model and iteratively adds the independent variable that contributes the most to the model's performance based on a predetermined criterion, such as a

significant p-value or an increase in the adjusted R-squared. This process continues until no more variables meet the inclusion criterion.

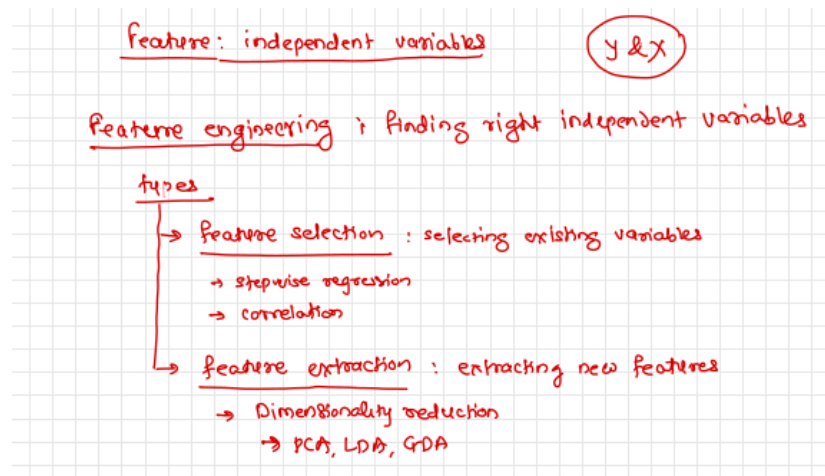
- Backward Elimination
 - Starts with a set of independent variables.
 - Deletes one at a time.
 - Then tests to see if the removed variable is statistically significant.
 - In backward elimination, the algorithm starts with a model that includes all independent variables and iteratively removes the least significant variable based on a predetermined criterion, such as a non-significant p-value or a decrease in the adjusted R-squared. This process continues until no more variables meet the exclusion criterion.
- Bidirectional Elimination/Stepwise Selection
 - It is the combination of the first two methods that test which variables should be included or excluded.

▼ Advantages

- The ability to manage large amounts of potential predictor variables, fine-tuning the model to choose the best predictor variables from the available options.
- It's faster than other automatic model-selection methods.
- Watching the order in which variables are removed or added can provide valuable information about the quality of the predictor variables.

▼ Disadvantages

- Stepwise regression often has many potential predictor variables but too little data to estimate coefficients meaningfully. Adding more data does not help much, if at all.
- If two predictor variables in the model are highly correlated, only one may make it into the model (multi-collinearity).
- R-squared values are usually too high.
- Adjusted r-squared values might be high, and then dip sharply as the model progresses. If this happens, identify the variables that were added or removed when this happens and adjust the model.
- Predicted values and confidence intervals are too narrow.
- P-values are given that do not have the correct meaning.
- Regression coefficients are biased and coefficients for other variables are too high.
- Collinearity is usually a major issue. Excessive collinearity may cause the program to dump predictor variables into the model.
- Some variables (especially dummy variables) may be removed from the model, when they are deemed important to be included. These can be manually added back in.



▼ Ridge Regression

▼ Overview

- Ridge regression, also known as Tikhonov regularization, is a linear regression technique that addresses the issue of multi-collinearity (high correlation between independent variables) by introducing a regularization term.
- It is commonly used when there are many correlated variables in a data set.
- Ridge regression works by adding a penalty term to the ordinary least squares (OLS) cost function, which shrinks the coefficient estimates towards zero.
- The penalty term is determined by the regularization parameter (λ), which controls the amount of regularization applied.

▼ Working

- The Ridge Regression equation is similar to the ordinary linear regression equation, but with an additional penalty term:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_n * x_n + \lambda * (b_1^2 + b_2^2 + \dots + b_n^2)$$

where:

- y represents the dependent variable
- $b_0, b_1, b_2, \dots, b_n$ are the coefficients of the independent variables
- $x_1, x_2, \dots, x_n$ are the independent variables
- λ is the regularization parameter
- The ridge penalty term is a crucial component of ridge regression. It acts as a regularization term that shrinks the coefficients towards zero, reducing their magnitudes and mitigating the impact of multi-collinearity (high correlation) between the independent variables.
- The value of the regularization parameter λ controls the amount of shrinkage applied to the coefficients. A higher λ results in greater shrinkage and a more conservative model, while a lower λ allows for less shrinkage.

▼ Bias-Variance Trade-Off

- Ridge regression involves a bias-variance trade-off. By introducing the ridge penalty, the model reduces the variance and potential overfitting caused by highly correlated variables.
- However, the ridge penalty also introduces a small bias by shrinking the coefficients. The optimal choice of the regularization parameter λ needs to strike a balance between bias and variance, leading to a model that generalizes well to unseen data.

▼ Application

- Dealing with multicollinearity: Ridge regression is particularly useful when there are highly correlated independent variables, as it can mitigate the impact of multicollinearity and stabilize the coefficient estimates.
- Predictive modeling: Ridge regression can be applied in various predictive modeling tasks, such as predicting housing prices, customer churn, or stock market trends, where there are multiple correlated predictors.

▼ Advantages

- Handles multicollinearity: Ridge regression is effective in reducing the impact of multicollinearity and stabilizing the coefficient estimates, preventing overfitting caused by highly correlated variables.
- Improves model robustness: By introducing regularization, ridge regression reduces the sensitivity of the model to outliers and noise in the data, leading to more robust predictions.
- Preserves variable information: Unlike variable selection techniques, ridge regression does not completely eliminate variables but instead shrinks their coefficients, allowing for the retention of valuable information from correlated predictors.

▼ Disadvantages

- Limited interpretability: Ridge regression does not result in sparse models, meaning that all variables are retained in the model with nonzero coefficients. This can make interpretation challenging compared to techniques that perform variable selection.
- Bias-variance trade-off: While ridge regression reduces variance and overfitting, it introduces a small bias by shrinking the coefficients. The optimal choice of the regularization parameter (λ) needs to strike a balance between bias and variance.
- Requires tuning of the regularization parameter: The choice of the regularization parameter λ must be carefully selected through techniques like cross-validation to optimize model performance. Improper tuning can lead to suboptimal results.
- Assumes linear relationship: Ridge regression assumes a linear relationship between the dependent variable and the independent variables. If the relationship is highly non-linear, alternative techniques may be more appropriate.
- Computational complexity: Ridge regression involves solving a modified cost function, which can be computationally more expensive than ordinary linear

regression, especially for large datasets.

▼ Lasso Regression

	Ridge Regression	Lasso Regression
Regularization Term	$\lambda * (b_1^2 + b_2^2 + \dots + b_n^2)$	$\lambda * (b_1 + b_2 + \dots + b_n)$
Shrinkage Effect	Shrinks coefficients towards zero	Can shrink coefficients to exactly zero
Feature Selection	Does not perform feature selection	Can perform feature selection by setting some coefficients to zero
Number of Selected Features	Retains all features with reduced magnitude	May select a subset of important features
Handling Multicollinearity	Reduces the impact of multicollinearity by shrinking coefficients	Can completely eliminate correlated features by setting their coefficients to zero
Model Interpretability	Coefficients retain their signs and relative magnitudes	Coefficients can be zero or non-zero, affecting interpretability
Computational Complexity	Computationally efficient	Can be computationally more expensive, especially with a large number of features
Suitable for	Dealing with multicollinearity	Feature selection and sparse models

▼ Elastic Net Regression

▼ Overview

- Elastic Net regression is a regularization technique that combines both L1 (Lasso) and L2 (Ridge) regularization penalties.
- It is used to handle situations where there are many correlated independent variables and to perform feature selection.
- The Elastic Net regression model strikes a balance between the sparsity-inducing L1 penalty and the coefficient shrinkage of the L2 penalty.

▼ Working

- Elastic Net regression adds a regularization term to the ordinary least squares (OLS) cost function that includes both L1 and L2 penalties.
- The Elastic Net regression equation can be represented as:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_n * x_n + \lambda_1 * (|b_1| + |b_2| + \dots + |b_n|) + \lambda_2 * (b_1^2 + b_2^2 + \dots + b_n^2)$$

- In this equation, y represents the dependent variable, and $b_0, b_1, b_2, \dots, b_n$ are the coefficients of the independent variables $x_1, x_2, \dots, x_n$, respectively. λ_1 and λ_2 are the regularization parameters that control the amount of L1 and L2 regularization applied.
- The optimization process in Elastic Net regression aims to minimize the combined error of the OLS cost function and the regularization term by adjusting the coefficients.

▼ Application

- Elastic Net regression is widely used in various domains, including finance, genetics, image processing, and natural language processing.

- It is commonly applied in situations where there are many predictors with correlated features, such as gene expression analysis, stock market prediction, or text classification.
- Elastic Net regression is particularly useful when there is a need for both feature selection and coefficient shrinkage, making it a versatile technique in predictive modeling.

▼ Advantages

- Handles multi-collinearity: Elastic Net regression can effectively handle multicollinearity by performing feature selection and shrinking coefficients towards zero.
- Feature selection: It can automatically select relevant features by setting some coefficients to zero, allowing for model simplification and improved interpretability.
- Robustness: Elastic Net regression offers improved robustness compared to individual L1 or L2 regularization techniques, as it combines their strengths.
- Generalization: It can provide better generalization performance by controlling both bias and variance.

▼ Disadvantages

- Computational complexity: Elastic Net regression can be computationally more expensive than individual L1 or L2 regularization due to the combined penalty term.
- Model interpretability: While Elastic Net regression performs feature selection, it may retain more features compared to Lasso regression, which can affect model interpretability.
- Selection of regularization parameters: The choice of the regularization parameters (λ_1 and λ_2) requires careful tuning, which can be challenging and may require cross-validation.



For Lasso and Ridge Regression video, click [here](#).

Evaluation Metrics

▼ MAE

- Definition: The Mean Absolute Error (MAE) is calculated as the average of the absolute differences between the predicted and actual values of the target variable across all data points.
- Calculation: To compute the MAE, you take the absolute difference between each predicted value and its corresponding actual value, sum them up, and then divide by the total number of data points.
- Interpretation: MAE represents the average magnitude of the errors made by the model. It gives you an idea of how close, on average, the predictions are to the actual values. The lower the MAE, the better the model's performance.
- Advantages: MAE is easy to understand, as it provides a straightforward measure of the average error. It is not sensitive to outliers, as it only considers the

absolute differences. MAE also preserves the scale of the target variable, making it more interpretable.

- Disadvantages: MAE does not differentiate between overestimation and underestimation. It treats all errors equally, regardless of their direction. In some cases, this may not capture the full impact of certain errors, especially if overestimation and underestimation have different consequences.
- Use Case: MAE is commonly used in various regression tasks, such as predicting housing prices, stock market values, or customer demand. It helps evaluate the accuracy and performance of the regression model by quantifying the average prediction error.
- Comparison with other metrics: MAE is similar to other error metrics like MSE (Mean Squared Error), but MSE squares the errors, which places more emphasis on larger errors. If interpretability is a priority, MAE may be preferred over MSE.

▼ MSE

- Definition: The Mean Absolute Error (MAE) is calculated as the average of the absolute differences between the predicted and actual values of the target variable across all data points.
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▼ RMSE

- Definition: RMSE is calculated as the square root of the average of the squared differences between the predicted and actual values of the target variable.
- Calculation: To compute RMSE, you first calculate the MSE by taking the average of the squared differences between predicted and actual values. Then, you take the

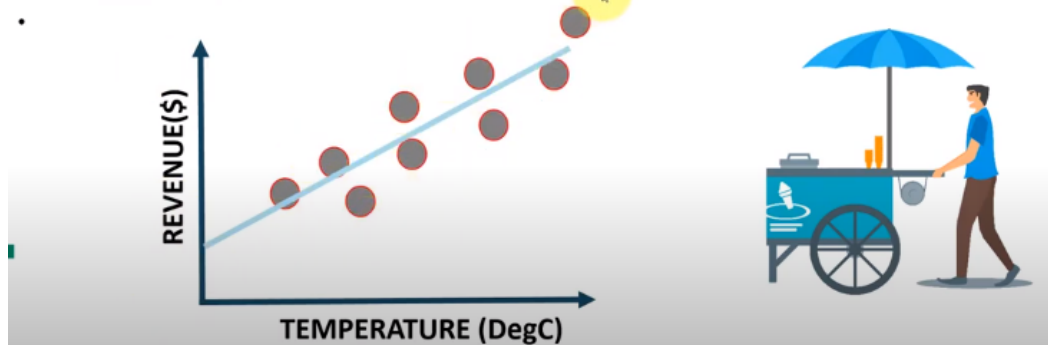
square root of the MSE to obtain RMSE.

- Interpretation: RMSE represents the average magnitude of the errors made by the model, similar to MAE and MSE. However, RMSE is expressed in the same unit as the target variable, making it more interpretable.
- Advantages: RMSE shares the advantages of MSE, as it rewards models for accurate predictions and is differentiable for optimization algorithms. RMSE also provides a measure of the average error in the original scale of the target variable.
- Disadvantages: Like MSE, RMSE is sensitive to outliers due to squaring the errors. Outliers can have a significant impact on the overall metric and may not accurately reflect the model's performance.
- Use Case: RMSE is commonly used in regression tasks when it is important to understand the average error in the original unit of the target variable. It helps assess the performance of the model and compare it with other models or benchmarks.
- Comparison with other metrics: RMSE is closely related to MSE but is expressed in the original unit of the target variable. It provides a more interpretable measure of the average error compared to MSE.

▼ R2

- Definition: R-squared (R^2) is a statistical metric used in regression analysis to measure the proportion of the variance in the dependent variable that can be explained by the independent variables.

- If $R^2 = 80$, this means that 80% of the increase in ice cream cart revenue is due to increase in temperature.



- Calculation:

$$R^2 = 1 - (SSE/SST)$$

$$R^2 = SSR/SST$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

- R2 is the coefficient of determination, representing the proportion of variance explained by the model.
- SSE (Sum of Squares of Residuals/Errors) is the sum of the squared differences between the actual values and the predicted values of the dependent variable.
- SST (Total Sum of Squares) is the sum of the squared differences between the actual values and the mean of the dependent variable.

$$y = (y_m - y_p) / (y_m - y_a)$$

- Interpretation: R2 ranges from 0 to 1, where 0 indicates that the independent variables do not explain any of the variance in the dependent variable, and 1 indicates that the independent variables explain all the variance in the dependent variable. A higher R2 value indicates a better fit of the regression model to the data.
- Advantages: R2 is widely used as a measure of the goodness of fit in regression models. It provides a standardized and interpretable metric to assess how well the model captures the variance in the dependent variable. R2 can be used to compare different models or evaluate the improvement in model performance when additional independent variables are added.
- Disadvantages: R2 has limitations, particularly when applied to complex models or in the presence of multicollinearity. R2 tends to increase with the addition of more independent variables, even if they are not truly contributing to the explanation of the dependent variable. Therefore, it is important to consider other metrics and evaluate the overall performance of the model.
- Use Case: R2 is commonly used in regression analysis to evaluate the predictive power and explanatory strength of the model. It is useful for assessing the effectiveness of the model in capturing the relationship between the independent and dependent variables.
- Comparison with other metrics: R2 is different from error metrics like MAE, MSE, and RMSE, as it focuses on the proportion of explained variance rather than the absolute error. R2 provides information about the model's goodness of fit, while error metrics provide insights into the magnitude of the errors made by the model.

▼ Adjusted R2

- Definition: Adjusted R-squared (R2) is a modified version of R-squared that takes into account the number of independent variables in the regression model. It penalizes the inclusion of irrelevant or redundant variables.
- Calculation: Adjusted R2 is calculated by adjusting the R2 value based on the number of independent variables (k) and the total number of observations (n). The formula for adjusted R2 is:

$$AdjustedR2 = 1 - [(1 - R2) * (n - 1) / (n - k - 1)],$$

- where,
R2 is the unadjusted R-squared,
n is the number of observations,
k is the number of independent variables.

$$R_{adj}^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - k - 1} \right]$$

where:

- N is the number of points in your data sample.
- K is the number of independent regressors, i.e. the number of variables in your model, excluding the constant.
- Interpretation: Adjusted R2 ranges from $-\infty$ to 1. A higher adjusted R2 value indicates a better fit of the regression model, taking into account the trade-off between model complexity (number of variables) and explanatory power. Adjusted R2 accounts for the sample size and the number of predictors in determining the goodness of fit.
- Advantages: Adjusted R2 addresses the issue of overfitting by penalizing the inclusion of unnecessary variables. It provides a more accurate assessment of the model's goodness of fit when comparing models with a different number of variables. Adjusted R2 helps in selecting the most appropriate model by considering both fit and complexity.
- Disadvantages: Adjusted R2 can be less intuitive to interpret compared to R2. It is a relative measure that depends on the number of variables and the sample size. Adjusted R2 tends to be lower than R2, especially when adding more variables to the model. It may not capture the full complexity of the relationship between variables and should be used in conjunction with other evaluation metrics.
- Use Case: Adjusted R2 is commonly used when comparing regression models with different numbers of independent variables. It helps in model selection and determining the most parsimonious and effective model.
- Comparison with other metrics: Adjusted R2 is similar to R2 but accounts for the number of variables in the model. It provides a more conservative estimate of the model's explanatory power by adjusting for the complexity of the model.



Correlation and R-squared are related measures but capture different aspects of the relationship between variables. Correlation quantifies the strength and direction of the linear relationship, while R2 represents the proportion of variance explained by the independent variables in a regression model.



MSE $\rightarrow$ SSE/N $\rightarrow$ Related to Variance as it is also squared.
RMSE (Standard Error) $\rightarrow$ root(SSE/N) $\rightarrow$ Related to Standard Deviation.

▼ 3.2. Classification

Overview

- Classification is a type of supervised learning in machine learning that involves assigning a label or category to a set of data based on its features or characteristics.

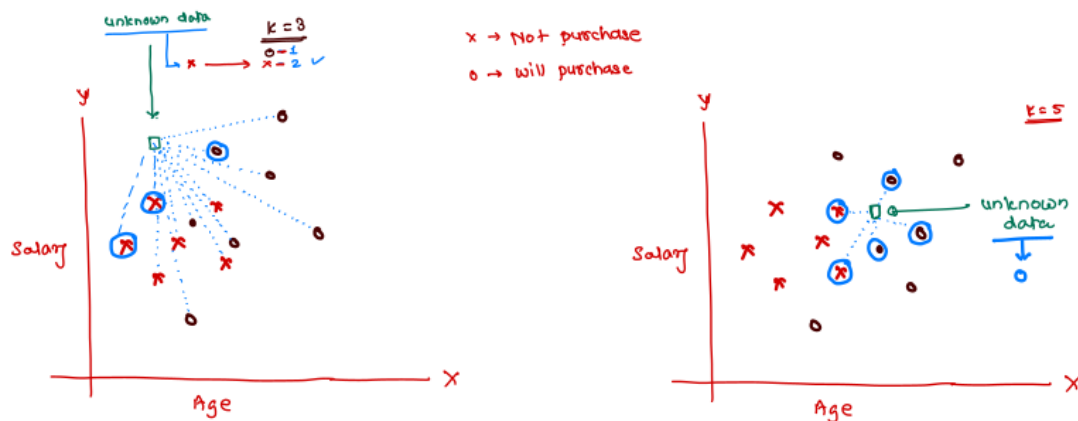
- In classification, the data is divided into two or more classes based on the characteristics or features of the data. The machine learning model learns to classify new or unseen data based on the patterns or relationships in the labelled training data.

Types of Classification

▼ K-Nearest Neighbour

▼ Overview

- KNN is a non-parametric algorithm that makes predictions based on the similarity of input data points to the data points in the training set.
- It operates on the principle that similar data points tend to have similar labels or outcomes.
- It can be used to solve both, classification and regression problems. However, it is more widely used to solve classification problems.



▼ Working

- KNN stores the entire training data set with labeled data points and their corresponding classes or values.
- When a new unlabeled data point is given for prediction, KNN starts by calculating the distance between the new point and all the data points in the training set.
- The distance metric used is typically Euclidean distance, but other metrics like Manhattan distance or Minkowski distance can also be employed.
- KNN selects the k nearest neighbors to the new data point based on the calculated distances. The value of k is a user-defined parameter.
- For classification tasks, KNN assigns the class label to the new data point based on the majority class among its k nearest neighbors. The class labels of the neighbors are determined by a voting scheme.
- For regression tasks, KNN calculates the average or weighted average of the values of its k nearest neighbors and assigns that as the predicted value for the new data point.

- In the case of classification, ties can occur when multiple classes have an equal number of neighbors. Different tie-breaking strategies can be employed to handle such situations.
- KNN does not involve an explicit training phase or model building. Instead, it stores the entire training dataset and uses it during the prediction phase.
- The choice of the value of k is crucial and can significantly affect the model's performance. A small k value may lead to overfitting, while a large k value may result in oversimplification.
- Feature scaling is often necessary in KNN since it is distance-based. Normalizing or standardizing the features can ensure that no single feature dominates the distance calculation.
- KNN is a lazy learning algorithm, as it does not generalize or extract explicit knowledge from the training data. It makes predictions based on local patterns observed in the training dataset.

▼ Application

- KNN is commonly used in pattern recognition, image classification, and computer vision tasks.
- It is useful in recommendation systems for collaborative filtering.
- KNN can be applied in anomaly detection and clustering analysis.

▼ Advantages

- KNN is simple and easy to understand.
- It does not make assumptions about the underlying data distribution.
- It can handle both classification and regression tasks.
- KNN is a lazy learning algorithm, as it does not involve an explicit training phase.

▼ Disadvantages

- KNN can be computationally expensive, especially for large datasets, as it requires calculating distances to all training samples.
- The choice of the value of k can impact the model's performance.
- KNN is sensitive to the scale of the features, and feature normalization may be necessary.
- It may struggle with high-dimensional data, as the concept of distance becomes less meaningful in higher dimensions.

▼ Logistic Regression



Why not use Linear Regression for Binary Classification instead of Logistic Regression?

1. The best fit line we get in linear regression is susceptible to outliers present in the data set which affects the output of our model.
2. If we use Linear Regression, we are working on the assumption that the data is linear which may result in a not-so-good model if the data isn't linear. Logistic Regression, on the other hand, is capable of handling non-linear data.
3. Logistic regression provides a probabilistic output and is appropriate for binary classification, whereas linear regression does not have a direct probabilistic interpretation.

▼ Overview

- Logistic Regression is a supervised machine learning algorithm used for binary classification tasks. It is designed to predict the probability of an event or the likelihood of a binary outcome based on input features.
- Unlike linear regression, logistic regression can directly predict probabilities.
- Furthermore, those probabilities are well-calibrated when compared to the probabilities predicted by some other classifiers.

▼ Assumptions

- Target Variable is binary.
- Only meaningful variables are included.
- Model should have little to no multi-collinearity.
- The independent variables are linearly related to the log odds. This means, the data points should be linearly separable.
- We have large sample size.

▼ Working

▼ Application

▼ Advantages

▼ Disadvantages

▼ Support Vector Machine

▼ Overview

- Support Vector Machine is a powerful supervised machine learning algorithm used for both classification and regression tasks.
- It aims to find an optimal hyperplane that maximally separates the data points of different classes or fits the data with the smallest error margin.

▼ Working

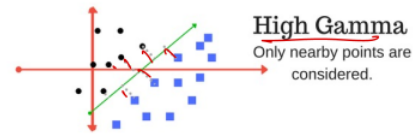
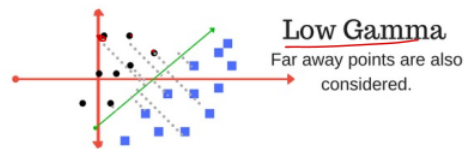
- SVM finds a hyperplane (decision boundary) in a high-dimensional feature space that maximizes the margin between classes.
- It selects a subset of training samples, called support vectors, that are closest to the decision boundary.
- The algorithm optimizes the hyperplane by minimizing the classification error or maximizing the margin using techniques like the hinge loss or quadratic programming.
- For nonlinearly separable data, SVM can use kernel functions (e.g., polynomial, radial basis function) to map the data into a higher-dimensional space, where linear separation is possible.
- During the prediction phase, SVM assigns new instances to a class based on which side of the decision boundary they fall on.

▼ Tuning Parameters

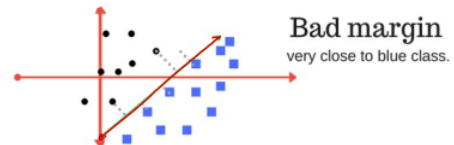
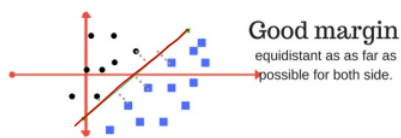
- Kernels
 - The learning of the hyperplane in linear SVM is done by transforming the problem using some linear algebra. This is where the kernel plays role.
- Regularization
 - The Regularization parameter tells the SVM optimization how much you want to avoid misclassifying each training example.
 - For large values of C , the optimization will choose a smaller-margin hyperplane if that hyperplane does a better job of getting all the training points classified correctly.
 - Conversely, a very small value of C will cause the optimizer to look for a larger margin separating hyperplane, even if that hyperplane misclassifies more points.



- Gamma
 - The gamma parameter defines how far the influence of a single training example reaches, with low values meaning 'far' and high values meaning 'close'.
 - With low gamma, points far away from plausible separation line are considered in calculation for the separation line.
 - Where as high gamma means the points close to plausible line are considered in calculation.



- Margin
 - A margin is a separation of line to the closest class points.
 - A good margin is one where this separation is larger for both the classes.
 - The points it passes through is called Support Vectors.



▼ Application

- Image classification and object recognition.
- Text classification and sentiment analysis.
- Bioinformatics for protein structure prediction.
- Financial market analysis and stock price prediction.
- Handwriting recognition and digit classification.

▼ Advantages

- SVM can handle high-dimensional data and is effective even when the number of features is greater than the number of samples.
- It is less prone to over-fitting due to the margin maximization objective.
- SVM can handle both linearly separable and non-linearly separable data using kernel functions.

▼ Disadvantages

- SVM can be computationally expensive, especially when dealing with large datasets.
- It can be sensitive to the choice of kernel and its parameters, which need to be carefully tuned.
- SVM does not directly provide probability estimates, requiring additional techniques like Platt scaling or cross-validation.
- Interpretability of SVM models can be challenging when using complex kernels.



Here, we have regularization, which is similar to the regularization which is done by Lasso, Ridge or Elastic Net Regression. In all these cases, we intentionally increase the error to reduce variance and introduce bias so that the model does not overfit has rooms for misclassification. For better understanding, click [here](#).

- Maximal Margin Classifiers
 - Keeps max margin from line that the closest point from each category of classification.
- Support Vector Classifiers
 - Separates data using a soft margin which allows miscalculations to lower variance, increase bias and thus increase model performance not on training data but on unknown/test data.
- Support Vector Machines
 - Start from lower dimension and move to upper dimensions while trying to classify data using Support Vector Classifiers (Kernel Functions).

▼ Decision Tree

▼ Overview

- A Decision Tree is a popular supervised machine learning algorithm used for both classification and regression tasks.
- It builds a tree-like model of decisions and their possible consequences based on the features of the input data.
- A decision tree is a flowchart-like structure in which each internal node represents a “test” on an attribute (e.g. whether a coin flip comes up heads or tails), each branch represents the outcome of the test, and each leaf node represents a class label (decision taken after computing all attributes).
- The paths from root to leaf represent classification rules.
- Tree based methods empower predictive models with high accuracy, stability and ease of interpretation.
- Unlike linear models, they map non-linear relationships quite well.
- Decision Tree algorithms are referred to as CART (Classification and Regression Trees).

▼ Terminologies

- Root Node: It represents entire population or sample and this further gets divided into two or more homogeneous sets.
- Splitting: It is a process of dividing a node into two or more sub-nodes.
- Decision Node: When a sub-node splits into further sub-nodes, then it is called decision node.
- Leaf/ Terminal Node: Nodes do not split is called Leaf or Terminal node.
- Pruning: When we remove sub-nodes of a decision node, this process is called Pruning. You can say opposite process of splitting.
- Branch / Sub-Tree: A sub section of entire tree is called Branch or Sub-Tree.

- Parent and Child Node: A node, which is divided into sub-nodes is called Parent Node of Sub-Nodes whereas Sub-Nodes are the child of Parent Node.

▼ Working

- The algorithm starts with a root node and selects the best feature to split the data based on certain criteria, such as Information Gain or Gini impurity.
- The data set is partitioned into subsets based on the selected feature's values, creating child nodes.
- Subsets should be made in such a way that each subset contains data with the same value for an attribute.
- This splitting process is recursively applied to each child node until a stopping criterion is met, such as reaching a maximum depth or having a minimum number of instances in a node.
- The decision tree model can be represented as a series of if-else conditions, where each internal node represents a decision based on a feature, and each leaf node represents a class label or a predicted value.
- During the prediction phase, new instances traverse the decision tree from the root to a leaf node based on their feature values, and the corresponding class label or value is assigned.

▼ Assumptions

- At the beginning, the whole training set is considered as the root.
- Feature values are preferred to be categorical. If the values are continuous, they are discretized prior to building the model.
- Records are distributed recursively on the basis of attribute values.

▼ Types

- Categorical Variable Decision Tree
 - Decision Tree which has categorical target variable (Classification).
- Continuous Variable Decision Tree
 - Decision Tree that has continuous target variable (Regression).

▼ Application

- It is one of the more popular classification algorithms being used in Data Mining.
- Determination of likely buyers of a product using demographic data to enable targeting of limited advertisement budget.
- Help with prioritisation of emergency room patient treatment using a predictive model based on factors such as age, blood pressure, gender, location etc.

▼ Advantages

- Easy to understand.
- Useful in data exploration.
- Decision trees implicitly perform variable screening or feature selection.
- Less effort for data preparation.

- Less cleaning required and also not affected by outliers.
- Data type is not a constraint.
- Decision Tree is considered to be a non-parametric method.
- The number of hyper-parameters to be tuned is almost null.

▼ Disadvantages

- Overfitting
- Not fit for continuous variables.
- Decision trees can be unstable because small variations in the data might result in a completely different tree being generated. This is called variance, which needs to be lowered by methods like bagging and boosting.
- Decision tree learners create biased trees if some classes dominate. It is therefore recommended to balance the data set prior to fitting with the decision tree.
- Calculations can become complex when there are many class label.

▼ Gini Index

The Gini index, also known as the Gini impurity or Gini coefficient, is a measure of impurity or inequality often used in decision tree-based machine learning algorithms for assessing the quality of a split during the tree-building process. It quantifies the extent of the class distribution imbalance within a set of data.

The Gini index is calculated as follows:

1. Calculate the probability (p) of each class within the set of data being evaluated.
2. Square each probability (p^2) for each class.
3. Sum up the squared probabilities for all classes.
4. Subtract the sum from 1.

The resulting value ranges from 0 to 1, where 0 indicates perfect purity (all instances belong to the same class) and 1 indicates maximum impurity (the classes are evenly distributed).

In decision tree algorithms, the Gini index is used to evaluate candidate splits. When choosing a split point, the algorithm aims to minimize the Gini index value, as this corresponds to a more homogeneous distribution of classes within the resulting subsets.

▼ Information Gain

Information gain is a measure used in decision tree algorithms, such as ID3 (Iterative Dichotomiser 3) and C4.5, to evaluate the usefulness of a feature in classifying or predicting the outcome variable.

Information gain quantifies the reduction in entropy or disorder in the dataset achieved by splitting the data based on a particular feature. It measures how much information a feature provides about the class labels or target variable.

The steps involved in calculating information gain are as follows:

1. Calculate the entropy of the original dataset, which represents the overall disorder or uncertainty of the class labels.
2. Split the dataset based on a feature and calculate the weighted average entropy of the resulting subsets. The weights are determined by the proportions of instances in each subset.
3. Calculate the information gain by subtracting the weighted average entropy of the subsets from the entropy of the original dataset.

The higher the information gain, the more valuable the feature is in terms of reducing the uncertainty about the class labels. Features with higher information gain are considered more informative and are preferred as splitting criteria in decision tree algorithms.

Information gain is based on the concept of entropy, which measures the impurity or disorder in a set of class labels. By selecting features that maximize information gain, decision tree algorithms can effectively split the data to create homogeneous subsets with respect to the class labels.



In case of Regression Tree, the value obtained by terminal nodes in the training data is the mean response of observation falling in that region. Thus, if an unseen data observation falls in that region, we'll make its prediction with mean value.

In case of Classification Tree, the value (class) obtained by terminal node in the training data is the mode of observations falling in that region. Thus, if an unseen data observation falls in that region, we'll make its prediction with mode value.

▼ Naive Bayes

Evaluation Metrics



For a balanced data set (upto 65-35% binary split), we use Accuracy metric.

For an imbalanced data set, we used F1-Score, Recall and Precision.

Click [here](#) to understand Classification Metrics.

▼ Confusion Matrix

- A confusion matrix is a table that summarizes the performance of a classification model by presenting the counts of various outcomes.
- It is particularly useful when evaluating the accuracy of a classification model, especially in scenarios where the classes are imbalanced.
- A confusion matrix typically has four components:
 - True Positives (TP)

- This is the count of instances that are actually positive (belonging to the positive class) and are correctly classified as positive by the model.
- For example, if we are building a model to detect spam emails, TP represents the count of actual spam emails that the model correctly classifies as spam.
- True Negatives (TN)
 - This is the count of instances that are actually negative (belonging to the negative class) and are correctly classified as negative by the model.
 - In the spam email example, TN represents the count of actual non-spam emails that the model correctly classifies as non-spam.
- False Positives (FP)
 - Also known as Type I errors, these are the instances that are actually negative but are incorrectly classified as positive by the model.
 - In the spam email example, FP represents the count of non-spam emails that the model wrongly classifies as spam.
- False Negatives (FN)
 - Also known as Type II errors, these are the instances that are actually positive but are incorrectly classified as negative by the model.
 - In the spam email example, FN represents the count of spam emails that the model mistakenly classifies as non-spam.

		Actual Conditions	
		Actual Condition Positive	Actual Condition Negative
Predicted Conditions	Total Population = P + N		
	Predicted Condition Positive	True Positive (TP)	False Positive (FP) Type 1 Error False Alarm, Overestimation
	Predicted Condition Negative	False Negative (FN) Type 2 Error Miss, Underestimation	True Negative (TN)

▼ Accuracy

- Classification accuracy is a commonly used evaluation metric that measures the overall correctness of a classification model.
- It is calculated as the ratio of the number of correctly classified instances to the total number of instances in the data set.
- The formula for classification accuracy is:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

		Actual Conditions	
		Actual Condition Positive	Actual Condition Negative
Predicted Conditions	Total Population = P + N		
	Predicted Condition Positive	True Positive (TP)	False Positive (FP) Type 1 Error False Alarm, Overestimation
	Predicted Condition Negative	False Negative (FN) Type 2 Error Miss, Underestimation	True Negative (TN)

▼ Recall

- Recall, also known as **Sensitivity** or **True Positive Rate**, is an evaluation metric that measures the ability of a classification model to correctly identify positive instances from the actual positive instances in the data set.
- The formula for recall is:

$$Recall = TP / (TP + FN)$$

		Actual Conditions	
		Actual Condition Positive	Actual Condition Negative
Predicted Conditions	Total Population = P + N		
	Predicted Condition Positive	True Positive (TP)	False Positive (FP) Type 1 Error False Alarm, Overestimation
	Predicted Condition Negative	False Negative (FN) Type 2 Error Miss, Underestimation	True Negative (TN)

Recall

$$\frac{TP}{TP + FN}$$

- The numerator of the recall formula represents the count of true positive instances, which are the correctly identified positive instances. The denominator (TP + FN) represents the total number of actual positive instances in the data-set.
- A higher recall value indicates a better ability of the model to identify positive instances accurately.
- Recall is particularly useful in scenarios where the cost of false negatives is high or when the emphasis is on minimizing the number of false negatives. For example, in medical diagnosis, recall is important to ensure that potentially critical cases are not missed.

		Actual Conditions	
		Cancer	Not Cancer
Predicted Conditions	Cancer	True Positive (TP)	False Positive (FP) Type 1 Error Important mail classified as Spam.
	Not Cancer	False Negative (FN) Type 2 Error Spam classified as Important Mail.	True Negative (TN)
		High Recall	

- However, it's important to note that optimizing for recall may lead to higher false positives (Type I errors). Therefore, it's crucial to consider the specific requirements and constraints of the classification problem and strike a balance between recall and other evaluation metrics such as precision, accuracy, or F1 score.

▼ Precision

- Precision, or **Positive Prediction Value**, is an evaluation metric that measures the proportion of correctly predicted positive instances out of the total instances predicted as positive by the classification model.
- It focuses on the model's ability to avoid false positives.
- The formula for precision is:

$$Precision = TP / (TP + FP)$$

		Actual Conditions	
		Actual Condition Positive	Actual Condition Negative
Predicted Conditions	Predicted Condition Positive	True Positive (TP)	False Positive (FP) Type 1 Error False Alarm, Overestimation
	Predicted Condition Negative	False Negative (FN) Type 2 Error Miss, Underestimation	True Negative (TN)
		Precision	

- Precision provides insights into the model's ability to avoid false positives, indicating how many of the predicted positive instances are actually positive.

		Actual Conditions		
		Spam	Not Spam	
Predicted Conditions	Spam	True Positive (TP)	False Positive (FP) Type 1 Error Important mail classified as Spam.	High Precision
	Not Spam	False Negative (FN) Type 2 Error Spam classified as Important Mail.	True Negative (TN)	

- Optimizing for high precision may result in higher false negatives (Type II errors), where actual positive instances are wrongly classified as negative.



Whenever your False Positive is much more important, blindly use Precision.

Whenever your False Negative is much more important, blindly use Recall.

▼ F-Beta Score (Sørensen-Dice coefficient or Dice similarity coefficient (DSC))

- F-beta score is an evaluation metric that combines precision and recall into a single value, providing a balanced measure of a classification model's performance.
- It allows you to adjust the importance given to precision and recall based on your specific needs by using a parameter called beta.
- The formula for the F-beta score is:

$$Fbeta = (1 + beta^2) * (precision * recall) / ((beta^2 * precision) + recall)$$

- Beta
 - Beta is a parameter that determines the weight given to precision versus recall in the F-beta score.
 - It allows you to adjust the balance based on your specific requirements.
 - A higher beta value gives more weight to Recall, while a lower beta value gives more weight to Precision.
 - When beta is set to 1, it becomes the F-1 score, which balances precision and recall equally.
- The F-beta score combines precision and recall by taking their harmonic mean, which emphasizes the lower value of the two. It provides a single metric that considers both precision and recall, offering a balanced evaluation of the model's performance.
- Commonly used values for beta are 1 (F-1 score, balanced), 0.5 (F-0.5 score, more weight on Precision), and 2 (F-2 score, more weight on Recall).

▼ Specificity

- Specificity, also known as the true negative rate, is an evaluation metric used in binary classification tasks. It measures the proportion of correctly predicted

negative instances out of the total actual negative instances.

- The formula for specificity is:

$$\text{Specificity} = TN / (TN + FP)$$

- Specificity provides insights into the model's ability to avoid false positives, indicating how many of the actual negative instances are correctly classified by the model. It is particularly important in scenarios where correctly identifying negative instances is crucial, such as medical diagnostics or fraud detection.
- Specificity is complementary to sensitivity (recall), which measures the model's ability to correctly identify positive instances. Together, they provide a more comprehensive understanding of a model's performance by considering both positive and negative classifications.

▼ ROC AUC

- ROC (Receiver Operating Characteristic) curve and AUC (Area Under the Curve) are evaluation metrics used to assess the performance of binary classification models.

- ROC Curve:

- The ROC curve is a graphical representation of the model's performance by plotting the true positive rate (sensitivity) against the false positive rate (**1 - specificity**) at various classification thresholds.

$$FPR = FP / (TN + FP)$$

- The x-axis of the ROC curve represents the false positive rate, which is the proportion of actual negative instances incorrectly classified as positive.
 - The y-axis of the ROC curve represents the true positive rate, which is the proportion of actual positive instances correctly classified as positive.
 - Each point on the ROC curve corresponds to a particular classification threshold, and the curve shows the trade-off between true positive rate and false positive rate.
 - The diagonal line from (0,0) to (1,1) represents random guessing, while an ideal classifier would have a ROC curve that reaches the top left corner (1,1) indicating a high true positive rate and a low false positive rate.
- AUC (Area Under the Curve):
 - AUC is a summary measure of the ROC curve, representing the area under the curve.
 - The AUC value ranges between 0 and 1, with a higher value indicating better discrimination between the positive and negative classes.
 - An AUC of 0.5 suggests that the classifier performs no better than random guessing, while an AUC of 1 represents a perfect classifier with no false positives and a true positive rate of 1.
 - AUC provides a single scalar value that can be used to compare and evaluate different models.

4. Unsupervised Learning

- Unsupervised learning is a type of machine learning where the algorithm learns patterns and relationships from unlabeled (no dependent variable) data without explicit guidance or predefined target labels.
- In unsupervised learning, the objective is to explore the underlying structure or patterns within the data.
- The main goal of unsupervised learning is to discover interesting and meaningful patterns, groupings, or relationships within the data without any prior knowledge of the outcomes.
- It is of 3 types:
 - Clustering
 - Associations Rule Mining
 - Dimensionality Reduction

▼ 3.1. Clustering

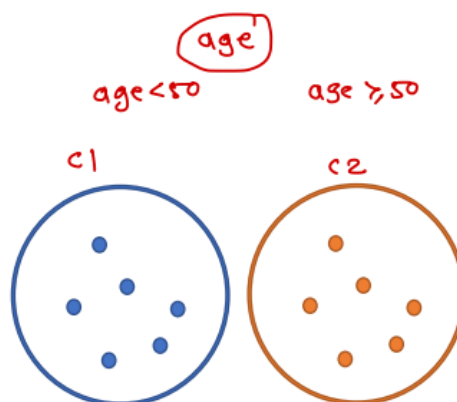
Overview

- Clustering is a technique used to identify underlying structures or relationships in the data without any prior knowledge or labels.
- These structures or relationships can be based on measures such as Euclidean-Based Distance, Correlation-Based Distance, and so on.

Types of Clusters

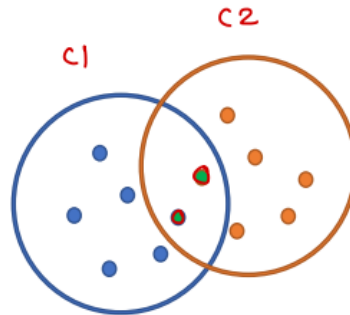
▼ Mutually Exclusive Clustering

- An item belongs exclusively to one cluster and not several
- Example: K-Means



▼ Overlapping Clustering

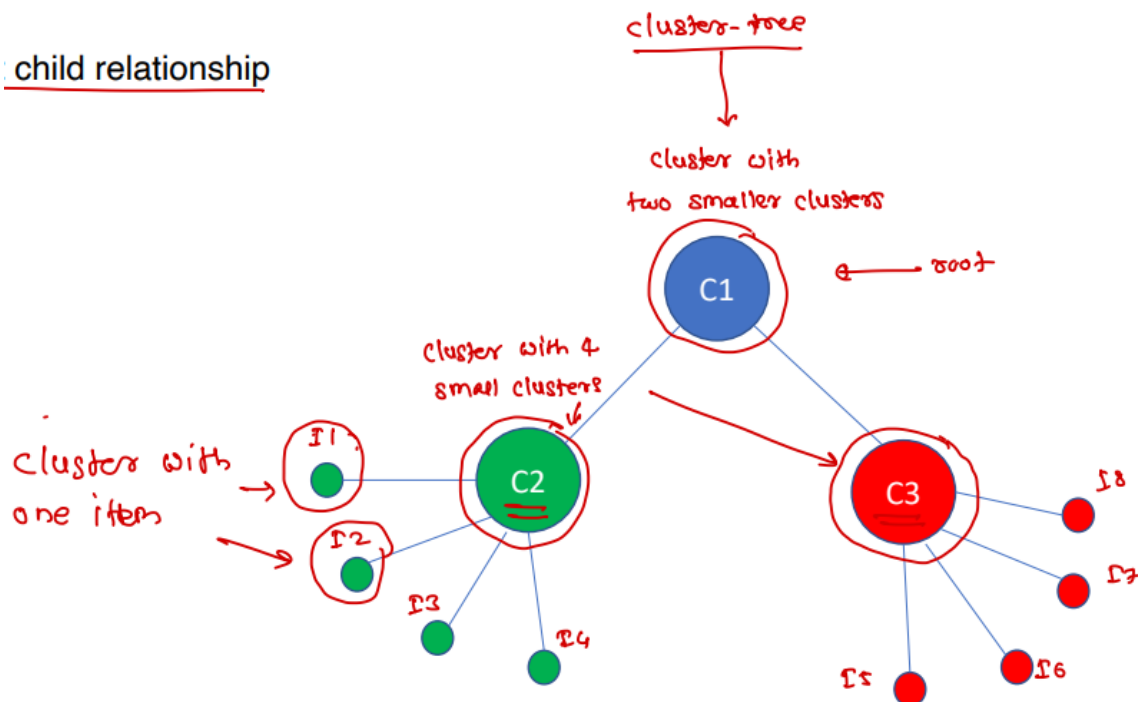
- An item can belong to multiple clusters.
- It's degree of association with each cluster is known.
- Example: Fuzzy/C-Means Clustering



▼ Hierarchical Clustering

- When 2 clusters have a parent-child relationship.
- It forms a tree like structure.
- Example: Hierarchical Clustering
- It can be represented using a Dendrogram

child relationship



▼ Probabilistic Clustering

Types of Clustering Algorithms

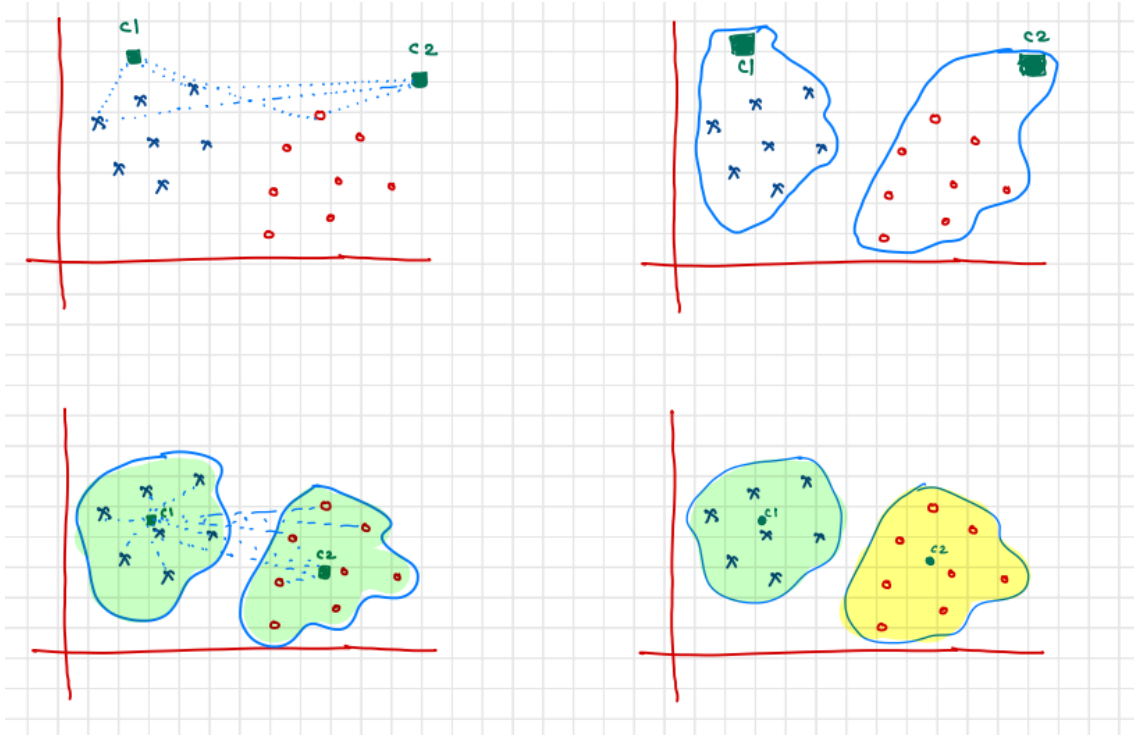
▼ K-Means

▼ Overview

- K-means is often referred to as **Lloyd's algorithm**.
- The K-Means algorithm clusters data by trying to separate samples in n groups of equal variance, minimising a criterion known as the *inertia* or Within-Cluster Sum-of-Squares (WCSS). This algorithm requires the number of clusters to be specified. It scales well to large numbers of samples
- This algorithm is an iterative algorithm that tries to partition the data-set into distinct non-overlapping subgroups (clusters), where data point belongs to only one group.
- It tries to make the inter-cluster data points as similar as possible while also keeping the clusters as different (far) as possible.
- It assigns data points to a cluster such that the sum of the squared distance between the data points and the cluster's centroid (arithmetic mean of all the data points that belong to that cluster) is at the minimum.
- The less variation we have within the clusters, the more homogeneous the data points are within the same cluster.

▼ How does it work?

1. Initialisation: Choose initial centroids randomly.
2. Cluster Assignment: Assign each data-point to its nearest centroid.
3. Move Centroid: Calculate a new centroid for that cluster. The centroid is simply the arithmetic mean of all the data points in the cluster.
4. Optimisation: Keep looping from step 2 until the older and the newer centroids in every cluster stop changing their positions.
5. Convergence: Finally, K-means clustering algorithm converges and divides the data points into two clusters clearly visible in multiple clusters.



▼ How to find the optimum number of clusters (K)?

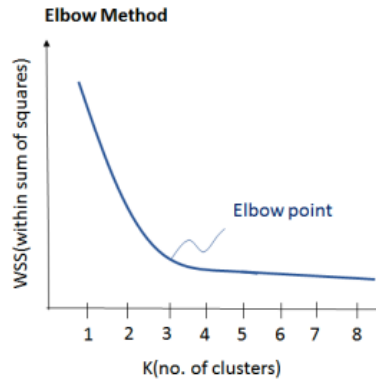
▼ Elbow Method

- Total within-cluster variance
 - Also known as Within Sum of Squares (WSS) → Sum of Squares of Distance between every point and cluster's centroid.
 - The sum of squared distances (Euclidean) between the items and the corresponding centroid.
 - Calculate WSS per cluster and add all of them.

$$\text{objective function} \leftarrow J = \sum_{j=1}^k \sum_{i=1}^n \underbrace{\|x_i^{(j)} - c_j\|^2}_{\text{Distance function}}$$

number of clusters number of cases case i centroid for cluster j

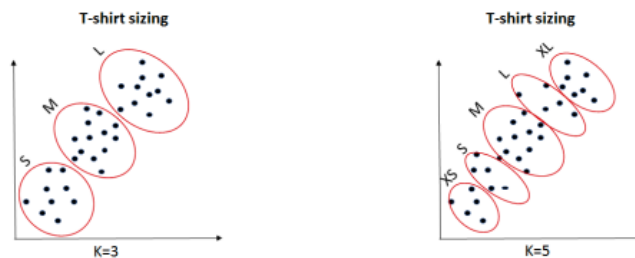
- Draw a curve between WSS and the number of clusters.
- It is called Elbow Method because the curve looks like a human arm and the Elbow point gives us the optimum number of clusters.
- inertia is the property which shows the WSS for all the clusters



Program for finding Elbow Point

▼ Purpose Method

- Get different clusters based on a variety of purposes.
- Partition the data on different metrics and see how well it performs for that particular case.



- K=3: If you want to provide only 3 sizes(S, M, L) so that prices are cheaper, you will divide the data set into 3 clusters.
- K=5: Now, if you want to provide more comfort and variety to your customers with more sizes (XS, S, M, L, XL), then you will divide the data set into 5 clusters.

▼ Application

- It is very popular and used in variety of applications such as Market Segmentation, Document Clustering, Image Segmentation (CNN) and Image Compression, etc.

▼ Advantages

▼ Disadvantages

- K-Means always makes spherical shape so if the shape of clusters gets much more complex, K-Means does a poor job in clustering the data.
- Doesn't let data point that are far-away from each other share the same cluster even though they obviously belong to the same cluster.



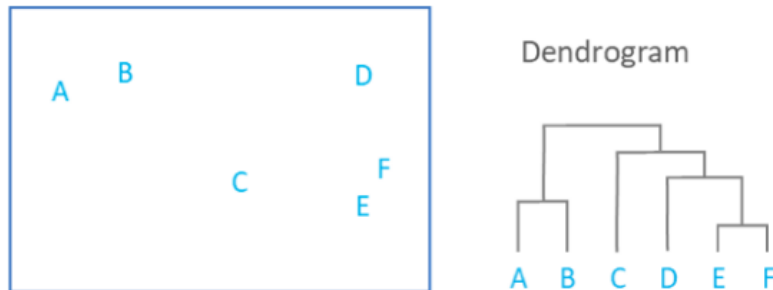
For K-Means Visualisation, click [here](#).

Program for K-Means

▼ Hierarchical Clustering

▼ Overview

- Hierarchical clustering, also known as Hierarchical Cluster Analysis (HCA), is an Unsupervised Clustering Algorithm which groups similar objects in a hierarchical manner.
- It involves dividing the data-set into smaller and smaller clusters or groups, where each group is nested inside the larger group until all individual data points are in their own cluster.
- It uses a distance measuring such as Euclidean distance (or the Manhattan distance) to determine the similarity between the data points or clusters.
- Hierarchical Clustering is usually accompanied by a **Dendrogram**. It indicates both, the similarity as well as the order that the clusters were formed.



▼ Types

▼ Agglomerative Clustering

▼ Overview

- It is a bottom-up approach in which each data point is initially considered as a separate cluster, and then the algorithm merges the two closest clusters into a single cluster.

▼ How does it work?

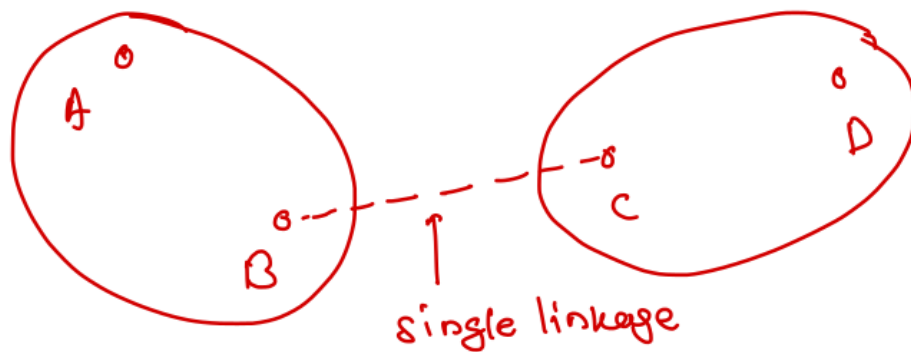
1. Start by assigning each item to a cluster. if you have N items, you now have N clusters, each containing just one item.
2. Find the closest (most similar) pair of clusters and merge them into a single cluster, so that now you have one less cluster.
3. Compute distances (similarities) between the new cluster and each of the old clusters.

4. Repeat steps 2 and 3 until all items are clustered into a single cluster of size N

▼ Step 3 can be done in different ways

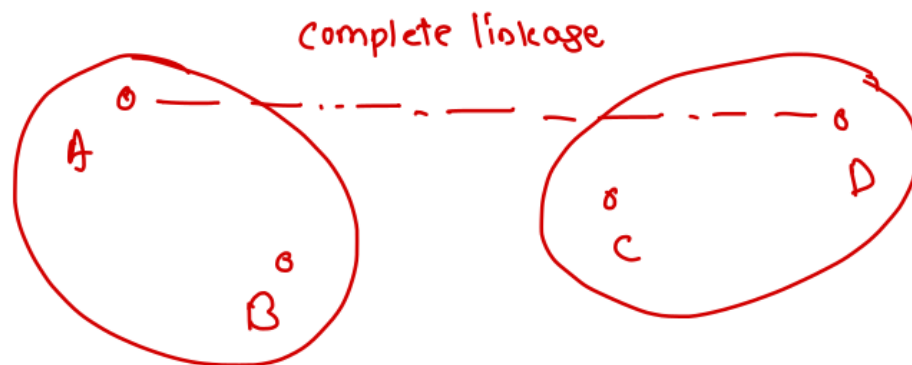
▼ Single-Linkage/Nearest Neighbour Clustering

- The distance between 2 groups is defined as the distance between their two closest members.
- It often yields clusters in which individuals are added sequentially to a single group.



▼ Complete Linkage/Farthest Neighbour Clustering

- The distance between 2 groups as the distance between their two farthest-apart members.



▼ Average Linkage/Unweighted Pair-Group Clustering

- Distance between two groups as the average distance between each of their members.
- (image)

▼ Divisive Clustering

- Also called as Top-Bottom approach.
- All data points start in a same cluster.

- Then using Parametric Clustering like K-Means divide the cluster into multiple clusters.
- For each cluster repeating the process find sub cluster till the desired number of clusters found.



How is K-Means clustering different from Hierarchical clustering?

K-Means clustering specifically tries to put the data into the number of clusters you tell it to while Hierarchical clustering just tells you, pairwise, what two things are most similar.

▼ 3.2. Associations Rule Mining

Overview

- Association rules mining is a data mining technique that involves finding correlations or associations between different items in a data-set.
- One of the most important applications of **Associative Rules Mining** is **Market Basket Analysis**. (Interview Question)

Types of Associations Rule Mining

▼ Apriori

▼ Overview

- This is given by R. Agrawal and R. Srikant in 1994 for finding frequent item-sets (products bought together) in a data-sets for boolean association rule.
- The name of the algorithm is Apriori because it uses prior knowledge of frequent item-set properties.
- The algorithm uses two steps to generate the frequent item-sets:
 - Candidate generation: The algorithm generates candidate item-sets of size k from the frequent item-sets of size k-1.
 - Pruning: The algorithm prunes the candidate item-sets that do not meet the minimum support threshold.
- Once the frequent item-sets are generated, association rules are generated by iterating over the frequent item-sets and generating all possible subsets. The confidence of each association rule is calculated and rules that meet the minimum confidence threshold are selected.
- To improve the efficiency of level-wise generation of frequent item-sets, an important property is used called Apriori property, which helps by reducing the search space.
- Apriori Property states that all non-empty subset of frequent item-set must be frequent.

▼ Terminologies

▼ Item-set

- It is a representation of the list of all items which form the association rule.
- Example: Itemset{Bread (**Antecedent**), Milk (**Consequent**)}

▼ Hyper-Parameters

- Used to tweak/optimize the performance of an algorithm.
- In this case we have 3:
 - Support
 - Confidence
 - Lift

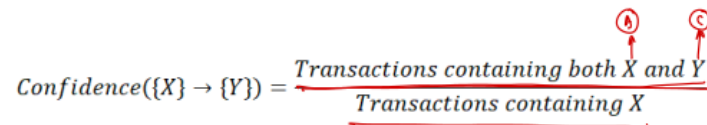
▼ Support

- This measure gives an idea of how frequent an item-set is in all the transactions.
- Mathematically, Support is the fraction of the total number of transaction in which item-set occurs.

$$\text{Support}(\{X\} \rightarrow \{Y\}) = \frac{\text{Transactions containing both } X \text{ and } Y}{\text{Total number of transactions}}$$

▼ Confidence

- This measure defines the likeliness of occurrence of consequent on the cart given that the cart already has the antecedents.
- Technically, Confidence is the conditional probability of occurrence of consequent given the antecedent.

$$\text{Confidence}(\{X\} \rightarrow \{Y\}) = \frac{\text{Transactions containing both } X \text{ and } Y}{\text{Transactions containing } X}$$


▼ Lift

- It controls the support (frequency) of consequent while calculating the conditional probability of occurrence of {Y} given {X}.
- Lift is the rise in probability of having {Y} on the cart with the knowledge of {X} being present over the probability of having {Y} on the cart without any knowledge about presence of {X}.

$$Lift = \frac{Support}{Supp(A) \times Supp(B)}$$

$$Lift(\{X\} \rightarrow \{Y\}) = \frac{\text{Confidence}}{\text{Fraction of transactions containing } Y}$$

$\downarrow$
 total no of transactions containing Y
 total no of transaction

▼ Disadvantages

- It may need to generate a huge number of candidate sets.
- It may need to repeatedly scan the database and check a large set-off candidates.

▼ Solved Sum

Transaction Id	Items Purchased
1	Bread, Cheese, Egg, Juice
2	Bread, Cheese, Juice
3	Bread, Milk, Yogurt
4	Bread, Juice, Milk
5	Cheese, Juice, Milk

Problem Statement

Item	Count	P(item)	Support
① Bread	4	4/16 = 0.25	4/5 = 0.8
② cheese	3	3/16	3/5 = 0.6
③ Egg	1	1/16	1/5 = 0.2
④ Juice	4	4/16	4/5 = 0.8
⑤ milk	3	3/16	3/5 = 0.6
⑥ Yogurt	1	1/16	1/5 = 0.2
	16		

Page 1 → Support

cons →	Bread	cheese	Juice	milk
Ant ↓				
<u>Bread</u> (4)	x	$\frac{2}{4} = 0.5^x$	$\frac{3}{4} = 0.75^{\checkmark}$	$\frac{2}{4} = 0.5^x$
<u>cheese</u> (3)	$\frac{2}{3} = 0.6^x$	x	$\frac{3}{3} = 1^{\checkmark}$	$\frac{1}{3} = 0.3^x$
<u>Juice</u> (4)	$\frac{3}{4} = 0.75^{\checkmark}$	$\frac{3}{4} = 0.75^{\checkmark}$	x	$\frac{2}{4} = 0.5^x$
<u>milk</u> (3)	$\frac{2}{3} = 0.6^x$	$\frac{1}{3} = 0.3^x$	$\frac{2}{3} = 0.6^x$	x

① { Bread → Juice } ④ { Juice → cheese }

② { cheese → Juice }

③ { Juice → Bread }

Page 2 → Confidence Matrix and Item-set formation

 For video explanation of Apriori, click [here](#).

Program for Apriori

- FP Trees
- ECLAT Algorithm

▼ 3.3. Dimensionality Reduction

Overview

- Dimensionality reduction is a technique used in machine learning and data analysis to reduce the number of input features or variables while retaining important information.
- It is employed to handle high-dimensional data, improve computational efficiency, and overcome the curse of dimensionality.
- It can be either of the following:
 - Feature Selection
 - Forward Selection
 - Backward Elimination
 - Feature Extraction
 - Principle Component Analysis (PCA)
 - Linear Discriminant Analysis (LDA)
 - Generalized Discriminant Analysis (GDA)

Types of Dimensionality Reduction Algorithms

- ▼ Principle Component Analysis

▼ Overview

- PCA transforms a set of correlated variables, known as features, into a new set of uncorrelated variables called principal components.
- These principal components are ranked in order of their importance, with the first component capturing the maximum variance in the data, the second component capturing the second maximum variance, and so on.
- By selecting a subset of the top principal components, the dimensionality of the data can be reduced.
- It works on a condition that while the data in a higher dimensional space is mapped to data in a lower dimension space, the variance of the data in the lower dimensional space should be maximum.
- This method was introduced by Karl Pearson.

▼ Working (Read Once)

1. Standardization

- The aim of this step is to standardize the range of the continuous initial variables so that each one of them contributes equally to the analysis.
- Once the standardization is done, all the variables will be transformed to the same scale.

2. Covariance Matrix Computation

- The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other.
- In other words, to see if there is any relationship between them as sometimes, variables are highly correlated in such a way that they contain redundant information.
- So, in order to identify these correlations, we compute the Covariance Matrix.
- The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables.
- Since the covariance of a variable with itself is its variance ($\text{Cov}(a, a) = \text{Var}(a)$), in the main diagonal (Top left to bottom right) we actually have the variances of each initial variable.
- Since the covariance is commutative ($\text{Cov}(a, b) = \text{Cov}(b, a)$), the entries of the covariance matrix are symmetric with respect to the main diagonal, which means that the upper and the lower triangular portions are equal.
- If positive then, the two variables increase or decrease together (correlated).
- If negative then, one increases when the other decreases (inversely correlated).

3. Compute Eigen Vectors and Values

- Eigenvectors and eigenvalues are the linear algebra concepts that we need to compute from the covariance matrix in order to determine the principal components of the data.
- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.
- These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.
- Organizing information in principal components this way, will allow you to reduce dimensionality without losing much information, and this by discarding the components with low information and considering the remaining components as your new variables.
- The principal components are less interpretable and don't have any real meaning since they are constructed as linear combinations of the initial variables.

4. Feature Vector

- Choose whether to keep all these components or discard those of lesser significance (of low eigenvalues), and form with the remaining ones a matrix of vectors that we call Feature Vector.
- Feature Vector is simply a matrix that has as columns the eigenvectors of the components that we decide to keep.

5. Ensemble Learning

- Ensemble is the art of combining diverse set of learners (individual models) together to improvise on the stability and predictive power of the model.
- Primarily used to improve the performance of a model.
- Other applications of ensemble learning include assigning a confidence to the decision made by the model, selecting optimal (or near optimal) features, data fusion, incremental learning, non-stationary learning and error-correcting.
- Types of Ensemble Learning are as follows:
 - Bagging
 - Random Forest
 - Boosting
 - Gradient Boost
 - Ada Boost
 - CAT Boost
 - XGBoost
 - Stacking

▼ Bagging (Bootstrap Aggregation)

- Bagging tries to implement similar learners on small sample populations and then takes a mean or mode of all the prediction.
- In generalized bagging, you can use different learners on different samples.
- This helps us to reduce variance error.

Types of Bagging Algorithms

▼ Random Forest

▼ Overview

- Random forest classifier creates a set of decision trees from randomly selected subset of training set.
- It then aggregates the votes from different decision trees to decide the final class of the test object.
- This works well because a single decision tree may be prone to a noise, but aggregate of many decision trees reduce the effect of noise giving more accurate results.

▼ Working

1. Random Subsampling: Random Forest randomly selects a subset of the training data with replacement (known as bootstrap sampling). This allows each tree to be trained on a different subset of the data, introducing variability and reducing overfitting.
2. Random Feature Selection: For each tree, Random Forest randomly selects a subset of features from the available features. This random feature selection helps to decorrelate the trees and ensure that different trees focus on different subsets of features.
3. Decision Tree Construction: Each tree in the Random Forest is constructed using a process similar to the construction of a regular decision tree. The tree is built by recursively partitioning the data based on the selected features and their split points, aiming to maximize the information gain or decrease in impurity.
4. Ensemble Prediction: Once all the trees are constructed, the Random Forest combines the predictions from each tree to make the final prediction. In classification tasks, the class with the majority vote is chosen as the prediction, while in regression tasks, the average of the predicted values is taken.

▼ Advantages

- Robustness: Random Forest is robust to noisy data and outliers due to the ensemble of multiple trees.
- High Accuracy: Random Forest generally achieves high accuracy by aggregating predictions from multiple trees, reducing the risk of overfitting.
- Feature Importance: Random Forest provides a measure of feature importance, indicating the contribution of each feature in the prediction process.

- Scalability: Random Forest can handle large datasets with high-dimensional feature spaces efficiently.

▼ Disadvantages

- Interpretability: The ensemble nature of Random Forest makes it less interpretable compared to individual decision trees.
- Computational Complexity: Training a Random Forest can be computationally expensive, especially for a large number of trees or complex datasets.
- Memory Usage: Random Forest requires memory to store multiple trees, which can be a limitation when working with very large datasets.

▼ Boosting

- Boosting refers to a family of algorithms that are able to convert weak learners to strong learners.
- Boosting is an iterative technique which adjust the weight of an observation based on the last classification.
- If an observation was classified incorrectly, it tries to increase the weight of this observation and vice-versa.
- Types of algorithms in Boosting are as follows:
 - Gradient Boosting
 - eXtreme Gradient Boosting
 - CAT Boost
 - AdaBoost

Types of Boosting

▼ Gradient Boosting Machines

▼ Overview

- Gradient Boosting Machines build an ensemble of weak prediction models (typically decision trees) sequentially, with each subsequent model learning from the mistakes of its predecessors.
- The models are added to the ensemble in a way that minimizes the overall prediction error.

▼ Working

1. Initialization: GBMs start by creating a simple initial model, often referred to as the base or weak learner. This can be a decision stump, which is a shallow decision tree with only a few levels.
2. Sequential Model Building: The subsequent models are built iteratively. Each new model is trained to predict the residuals (the differences between the actual target values and the predictions of the previous models). The model is fitted to minimize the residual error using a gradient descent algorithm.
3. Ensemble Aggregation: The predictions of all the models in the ensemble are combined to make the final prediction. The individual predictions are weighted based on their performance and contribution to the overall ensemble.

4. Learning Rate: GBMs often incorporate a learning rate, which controls the contribution of each model to the ensemble. A lower learning rate makes the ensemble more robust and helps prevent overfitting by reducing the impact of each individual model.

▼ Advantages

- High Predictive Accuracy: GBMs are known for their high accuracy and strong predictive performance. They can handle complex relationships between features and target variables.
- Flexibility: GBMs can handle a variety of data types, including numerical and categorical features.
- Feature Importance: GBMs provide insights into feature importance, helping to identify the most influential features in the prediction process.
- Handling Missing Data: GBMs can handle missing data by finding surrogate splits in decision trees.

▼ Disadvantages

- Computational Complexity: GBMs can be computationally expensive, especially when dealing with large datasets and complex models.
- Sensitivity to Outliers: GBMs are sensitive to outliers, and extreme values can have a disproportionate impact on the model.
- Potential Overfitting: GBMs have a higher risk of overfitting if the model complexity or number of iterations is not properly controlled. Regularization techniques such as early stopping or limiting the tree depth can help mitigate overfitting.

▼ Categorical Boosting

▼ Overview

- CAT Boost is an extension of gradient boosting that specifically addresses the challenges posed by categorical variables.
- It employs a novel algorithm that handles categorical features directly, without requiring pre-processing steps such as one-hot encoding or label encoding.
- This feature makes CAT Boost particularly useful when working with datasets that contain a mix of numerical and categorical features.

▼ Working

1. Categorical Feature Handling: CATBoost employs an innovative algorithm that naturally handles categorical features during the boosting process. It internally converts categorical features into numerical values using various techniques such as target encoding and combinations of mean and target encoding. This approach eliminates the need for explicit categorical feature encoding and retains the categorical nature of the data.
2. Gradient Boosting: CAT Boost builds an ensemble of weak prediction models, typically decision trees, in a similar manner to other gradient boosting frameworks. It iteratively adds trees to the ensemble, where each tree is trained to minimize the residual errors of the previous trees.

3. Ordered Boosting: CAT Boost incorporates an ordered boosting algorithm that improves the efficiency and accuracy of the gradient boosting process. It reduces the computation time and memory usage by applying a permutation-based technique to evaluate feature interactions.

▼ Advantages

- Automatic Handling of Categorical Features: CATBoost can handle categorical features directly, eliminating the need for manual encoding and reducing the dimensionality of the feature space.
- High Predictive Accuracy: CATBoost is known for its high accuracy and strong performance on a wide range of datasets.
- Efficient Implementation: CATBoost is optimized for efficiency and can handle large datasets and complex models effectively.
- Robust to Overfitting: CATBoost incorporates regularization techniques to prevent overfitting, such as gradient-based and leaf-wise regularization.

▼ Disadvantages

- Computational Complexity: Like other gradient boosting frameworks, CATBoost can be computationally expensive, especially with large datasets and complex models.
- Sensitivity to Hyperparameters: CATBoost has several hyperparameters that need to be tuned carefully to achieve optimal performance, which requires some experimentation.

▼ XGBoost

▼ Overview

- XGBoost is a decision-tree-based ensemble machine learning algorithm that uses a gradient boosting framework.
- XGBoost algorithm was developed as a research project at the University of Washington.
- XGBoost and Gradient Boosting Machines (GBMs) are both ensemble tree methods that apply the principle of boosting weak learners using the gradient descent architecture.
- However, XGBoost improves upon the base GBM framework through systems optimization and algorithmic enhancements.
- It aims to optimize both the predictive accuracy and computational efficiency of the model.
- XGBoost employs a combination of regularization techniques, advanced loss functions, and parallelized computation to achieve high performance.



▼ Working

- Parallelized Computation
 - XGBoost optimizes the computational efficiency by utilizing parallelized computing techniques.
 - It efficiently distributes the computation across multiple CPU cores and supports distributed computing on clusters to accelerate model training and prediction.
 - XGBoost approaches the process of sequential tree building using parallelized implementation.
- Tree Pruning:
 - XGBoost implements a tree pruning technique that removes unnecessary branches in decision trees during the building process.
 - This helps to control tree depth and reduce model complexity, enhancing generalization and preventing overfitting.
 - The stopping criterion for tree splitting within GBM framework is greedy in nature and depends on the negative loss criterion at the point of split.
 - XGBoost uses `max_depth` parameter as specified instead of criterion first, and starts pruning trees backward.
 - This 'depth-first' approach improves computational performance significantly.
- Hardware Optimization
 - This algorithm has been designed to make efficient use of hardware resources.
 - This is accomplished by cache awareness by allocating internal buffers in each thread to store gradient statistics.
 - Further enhancements such as 'out-of-core' computing optimize available disk space while handling big data-frames that do not fit into memory.
- Gradient-based Optimization:

- XGBoost uses gradient-based optimization to minimize a loss function, which quantifies the difference between the predicted and actual values.
- It iteratively adds new models to the ensemble, focusing on the examples with higher errors or residuals.

▼ Advantages

- Parallel Computing using OpenMP.
- Regularization.
- Enabled (internal) Cross-Validation.
- Handles missing values internally.
- Flexibility
- Availability
- Save and Reload.
- Tree Pruning.

▼ Disadvantages

▼ Stacking

- Stacking is an ensemble learning technique that combines multiple classification or regression models via a meta-classifier or a meta-regressor.
- The base level models are trained based on a complete training set, then the meta-model is trained on the outputs of the base level model as features.
- The base level often consists of different learning algorithms and therefore stacking ensembles are often heterogeneous.

Parameter	Confidence Interval Formula	Distribution	Critical Value	Standard Error
Population Mean	$\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$	Normal	z from Z-distribution	$\frac{\sigma}{\sqrt{n}}$
Population Proportion	$\hat{p} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	Normal (approximation)	z from Z-distribution	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Slope in Linear Regression	$b_1 \pm t \frac{s_{b1}}{\sqrt{SS_{xx}}}$	t-Distribution	t from t-distribution	$\frac{s}{\sqrt{SS_{xx}}}$

6. Neural Networks

Overview

- A neural network, also known as an artificial neural network (ANN), is a computational model inspired by the structure and function of biological neural networks in the human brain.
- It is a powerful machine learning algorithm used for a variety of tasks, including pattern recognition, regression, classification, and optimization.

- A neural network consists of interconnected nodes, called neurons or nodes, organized into layers.
- The input layer receives the input data, and the output layer produces the desired output.
- Between the input and output layers, there can be one or more hidden layers, which contain neurons that process and transform the data.

Working

1. Neuron Activation:

- Each neuron in the neural network receives inputs from the previous layer. These inputs are weighted by connection weights.
- The neuron applies an activation function to the weighted sum of the inputs and its bias.
- The activation function introduces non-linearity and determines the neuron's output, which is passed to the next layer.

2. Feedforward Propagation:

- The input data is fed into the input layer of the neural network.
- The input layer passes the data forward through the network, with each layer's outputs becoming inputs for the next layer.
- This process is known as feedforward propagation, as the data flows forward through the network.

3. Weighted Sum and Activation:

- In each layer, the neurons calculate a weighted sum of their inputs.
- The weighted sum is obtained by multiplying the input values by their corresponding connection weights and adding the bias term.
- The neuron then applies the activation function to the weighted sum, producing its output.

4. Activation Functions:

- Activation functions introduce non-linearities to the network, allowing it to learn complex patterns.
- Common activation functions include the sigmoid function, tanh function, and rectified linear unit (ReLU) function.
- Each activation function has different properties, such as different ranges and characteristics of non-linearity.

5. Backpropagation:

- Backpropagation is used to train the neural network by adjusting the weights and biases based on the error or loss between the predicted output and the actual output.
- The error is propagated backward through the network, starting from the output layer and moving towards the input layer.

- The gradients of the error with respect to the weights and biases are calculated using the chain rule of calculus.
- The weights and biases are then updated using an optimization algorithm, typically gradient descent, to minimize the error.

6. Training and Optimization:

- Neural networks are trained using labeled data, where the desired outputs are known.
- The training process involves iteratively presenting the training data to the network, adjusting the weights and biases, and minimizing the error.
- The optimization algorithm adjusts the weights and biases in the direction of steepest descent to reach the minimum of the error surface.
- The process continues until the network achieves satisfactory performance on the training data.

7. Testing and Prediction:

- Once the neural network is trained, it can be used for prediction on new, unseen data.
- The input data is fed forward through the network, and the output layer provides the predicted output or class probabilities.

Application

1. Image and Object Recognition: Neural networks, particularly convolutional neural networks (CNNs), have achieved remarkable success in tasks such as image classification, object detection, and facial recognition.
2. Natural Language Processing (NLP): Neural networks are used in NLP tasks such as language translation, sentiment analysis, text generation, and speech recognition. Recurrent neural networks (RNNs) and transformer models like the GPT series have been influential in this field.
3. Recommendation Systems: Neural networks are used in recommendation systems to provide personalized recommendations based on user preferences and behavior, improving user experience in areas like e-commerce, content streaming, and online advertising.
4. Time Series Analysis: Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks are utilized for time series analysis and forecasting, enabling applications such as stock market prediction, weather forecasting, and demand forecasting.
5. Autonomous Vehicles: Neural networks play a crucial role in autonomous vehicles for tasks like object detection, lane detection, and decision-making based on sensor data. Deep learning techniques have contributed significantly to advancements in self-driving car technology.
6. Medical Diagnosis: Neural networks are employed in medical imaging analysis, disease diagnosis, and prognosis prediction. They can analyze medical images, detect abnormalities, assist in diagnosis, and predict patient outcomes based on patient data.
7. Fraud Detection: Neural networks are used in fraud detection systems to identify suspicious patterns and anomalies in financial transactions, helping to prevent

fraudulent activities and protect against cybercrime.

8. **Robotics:** Neural networks contribute to various aspects of robotics, including robot perception, control, motion planning, and reinforcement learning. They enable robots to learn and adapt to their environment and perform complex tasks.
9. **Generative Models:** Generative adversarial networks (GANs) and variational autoencoders (VAEs) are used for generating realistic synthetic images, videos, and audio, as well as for data augmentation and synthesis.
10. **Gaming:** Neural networks have made significant contributions to the field of game playing. Deep reinforcement learning has been used to train agents to play complex games like chess, Go, and video games, achieving superhuman performance.

7. Deep Learning Algorithms

Deep learning algorithms are a subset of machine learning algorithms that are based on artificial neural networks with multiple layers (hence the term "deep"). These algorithms have revolutionized various fields, particularly computer vision and natural language processing. Here are some commonly used deep learning algorithms:

1. **Convolutional Neural Networks (CNNs):** CNNs are primarily used for image and video processing tasks. They consist of convolutional layers that extract features from input data, followed by pooling layers for dimensionality reduction, and fully connected layers for classification or regression.
2. **Recurrent Neural Networks (RNNs):** RNNs are designed to process sequential data, such as time series or text data. They use recurrent connections to pass information from previous steps to the current step, making them effective for tasks like speech recognition, language translation, and sentiment analysis.
3. **Long Short-Term Memory (LSTM) Networks:** LSTMs are a type of RNN that address the vanishing gradient problem. They can capture long-term dependencies in sequential data and have been widely used in speech recognition, natural language processing, and time series analysis.
4. **Generative Adversarial Networks (GANs):** GANs consist of a generator network and a discriminator network that are trained together in a competitive manner. GANs are used for generating realistic synthetic data, such as images, audio, and text, and have applications in image synthesis, style transfer, and data augmentation.
5. **Deep Reinforcement Learning:** Deep reinforcement learning combines deep learning with reinforcement learning, enabling agents to learn optimal actions in an environment through trial and error. Deep Q-Networks (DQNs) and policy gradient methods are common techniques used in this field, and they have achieved breakthrough results in game playing, robotics, and autonomous systems.
6. **Transformers:** Transformers are attention-based neural network architectures that have gained prominence in natural language processing tasks, such as machine translation and language understanding. The Transformer model, with its self-attention mechanism, has been the foundation for state-of-the-art models like BERT and GPT.
7. **Autoencoders:** Autoencoders are neural networks used for unsupervised learning and dimensionality reduction. They learn to encode the input data into a compressed representation and decode it back to its original form. Autoencoders have applications in data compression, anomaly detection, and feature extraction.

